

Effects of expanding higher education vacancies: effects on marginal and infra-marginal students

Preliminary Results

30 de septiembre de 2026

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Data and Sample

Data and sample

- We combine DEMRE and MINEDUC administrative records for admission cohorts 2007–2016.
- The resulting database links applications, enrollment outcomes, graduation outcomes, and PSU scores at the applicant level.
- The main analysis focuses on applicants within 25 points of the regular admission cutoff.

Statistic	Sample	N	Mean
Applications	Full sample	4,434,346	–
Unique students	Full sample	786,071	–
Program-years	Full sample	11,789	–
Running variable	BW25	839,108	-2.326
Above cutoff	BW25	839,108	0.433
Enrolls target	BW25	839,108	0.369
Enrolls university	BW25	839,108	0.833
Enrolls HE	BW25	839,108	0.870
Graduates target (8y)	Grad. sample	838,307	0.140
Graduates university (8y)	Grad. sample	838,307	0.402
Graduates HE (8y)	Grad. sample	838,307	0.423

RDD: Threshold crossing effects

RDD around admission cutoffs

- Running variable: application score centered around the regular admission cutoff,

$$r_{ipt} = score_{ipt} - cutoff_{pt}.$$

- Treatment indicator:

$$A_{ipt} = 1(r_{ipt} \geq 0).$$

- We compare applicants marginally above and below the cutoff within the same program-year.
- Main specification:

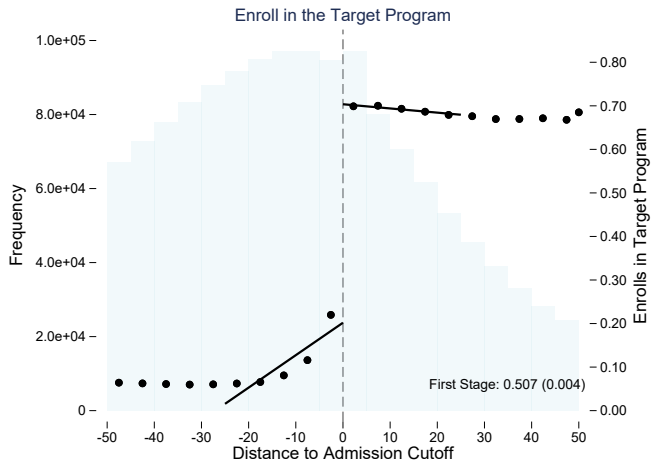
$$Y_{ipt} = \beta A_{ipt} + f(r_{ipt}) + \mu_{pt} + \varepsilon_{ipt}$$

Main bandwidth: $|r_{ipt}| \leq 25$. Standard errors are clustered at the program-year level.

RDD: Threshold crossing effects

Main Results

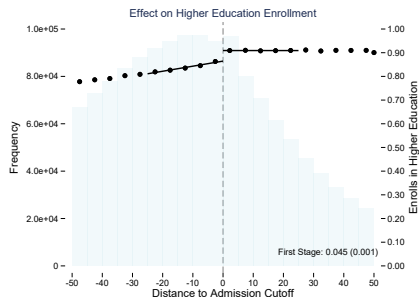
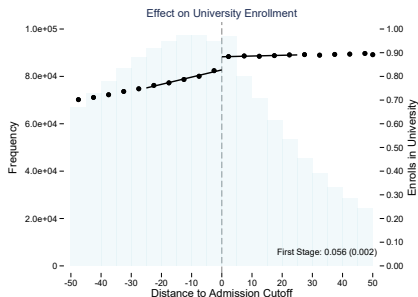
Enrollment in the Target Program



RDD estimate
0.507 (0.004)

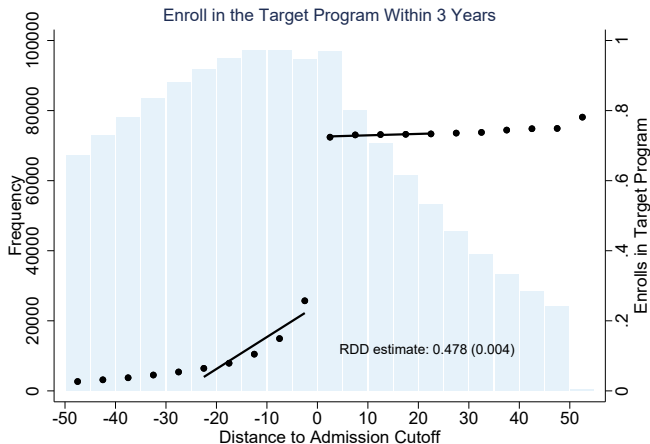
Crossing the cutoff
increases
target-program
enrollment by
approximately 51
percentage points.

Enrollment on University and HE Program



- Most applicants below the cutoff still access university or higher education through alternative options.
- Crossing the cutoff mainly changes **where students enroll**, rather than **whether they enroll**.

Target Enrollment Within Three Years

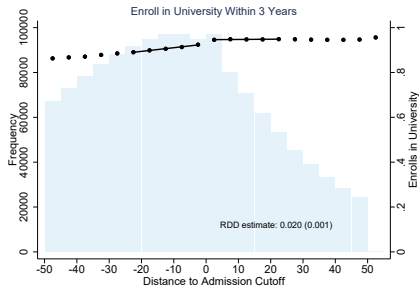


Regression BW = ± 25 . Plot range = ± 50 . FE: program-year. SE clustered by program-year. N=839108, clusters=8369.

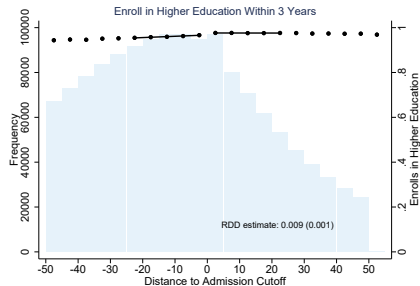
RDD estimate
0.478
(0.004)

Crossing the cutoff
increases
target-program
enrollment within
three years by
approximately 48
percentage points.

Broader Enrollment Outcomes Within Three Years



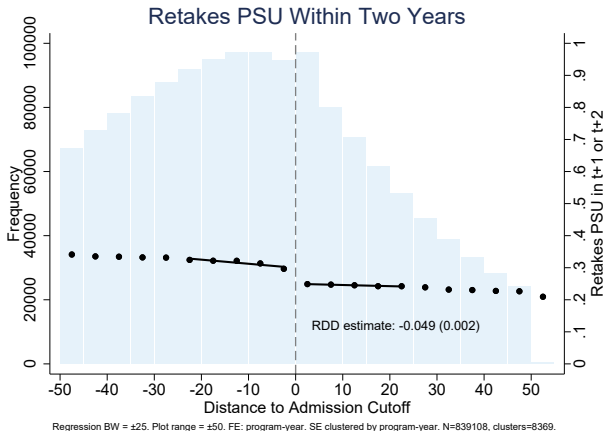
Regression BW = ± 25 . Plot range = ± 50 . FE: program-year. SE clustered by program-year. N=839108, clusters=8369.



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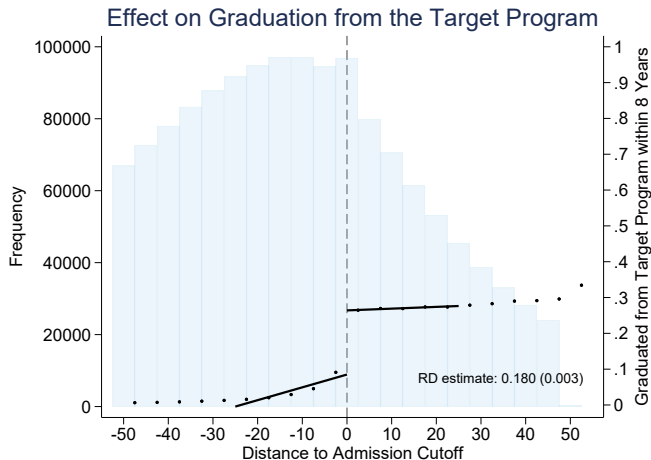
- Most applicants below the cutoff still access university or higher education through alternative options.
- The effect on overall higher-education participation remains small.
- The main effect continues to operate through the specific program attended.

PSU Retaking Within Two Years



- Crossing the cutoff reduces PSU retaking within the next two years.
- Applicants above the cutoff are less likely to re-enter the admission process.
- Admission eligibility lowers incentives to improve future admission outcomes.

Target Graduation Within Eight Years

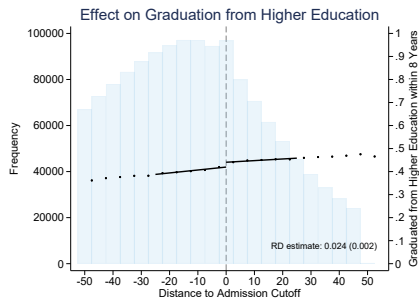
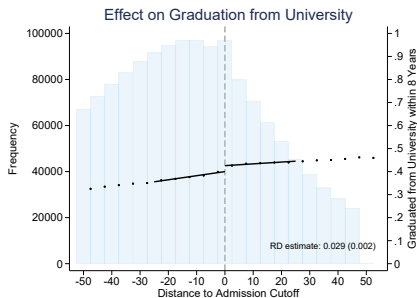


RDD estimate
0.180
(0.003)

Crossing the cutoff
increases graduation
from the target
program by
approximately 18
percentage points.

Graduation is measured within eight years of the application year.

Broader Graduation Outcomes Within Eight Years



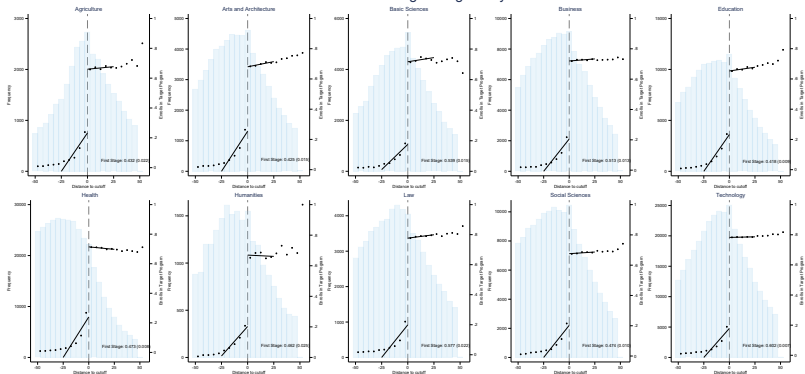
- Broader graduation effects are positive but much smaller than the target-program effect.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

RDD: Threshold crossing effects

Heterogeneity

Target Enrollment Effects by Field

RDD: Enroll in the Target Program by Field



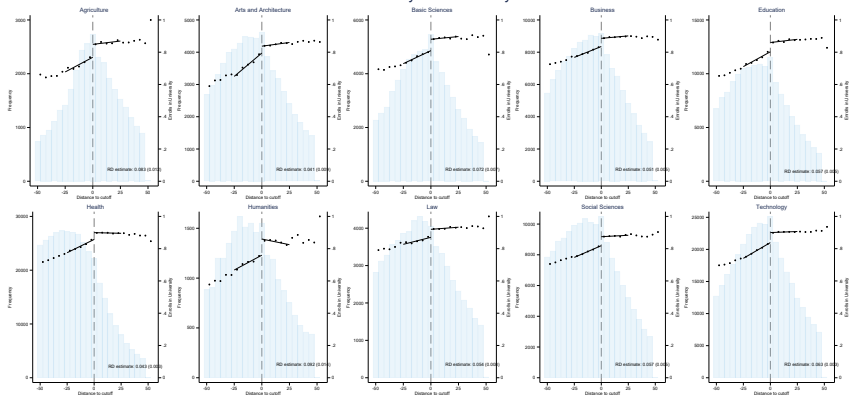
- Crossing the cutoff increases target-program enrollment in every field.
- Effects are largest in Technology, Law, Basic Sciences, and Business.

Target Enrollment Effects by Field

Field	Target	University	HE	N
Agriculture	0.432	0.083	0.065	21,102
Arts & Architecture	0.425	0.041	0.034	39,715
Basic Sciences	0.539	0.072	0.063	43,849
Business	0.513	0.051	0.034	79,964
Education	0.418	0.057	0.044	95,508
Health	0.473	0.043	0.039	206,050
Humanities	0.462	0.092	0.084	13,470
Law	0.577	0.054	0.054	37,000
Social Sciences	0.474	0.057	0.049	90,504
Technology	0.602	0.063	0.047	211,145

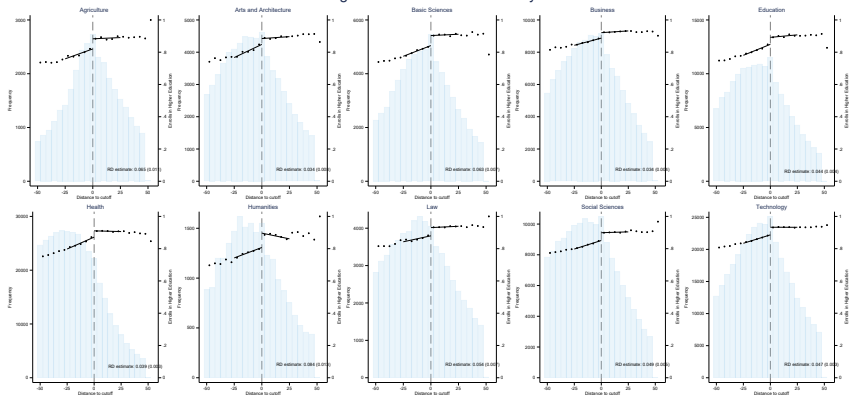
University enrollment effects by field.

RDD: University Enrollment by Field



Higher-education enrollment effects by field

RDD: Higher Education Enrollment by Field

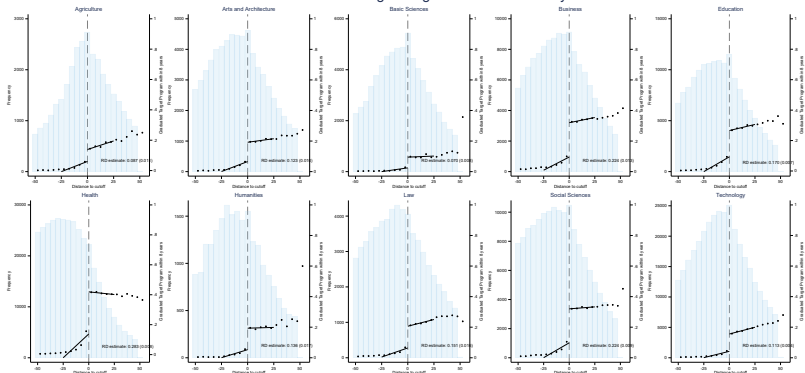


RDD estimates by field: university and HE enrollment

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	0.083	0.012	21,102	0.729
	HE	0.065	0.011	21,102	0.793
Arts & Architecture	University	0.041	0.009	39,715	0.723
	HE	0.034	0.008	39,715	0.808
Basic Sciences	University	0.072	0.007	43,849	0.771
	HE	0.063	0.007	43,849	0.809
Business	University	0.051	0.005	79,964	0.802
	HE	0.034	0.004	79,964	0.865
Education	University	0.057	0.005	95,508	0.757
	HE	0.044	0.004	95,508	0.815
Health	University	0.043	0.003	206,050	0.815
	HE	0.039	0.003	206,050	0.834
Humanities	University	0.092	0.014	13,470	0.713
	HE	0.084	0.013	13,470	0.775
Law	University	0.054	0.008	37,000	0.849
	HE	0.054	0.007	37,000	0.862
Social Sciences	University	0.057	0.005	90,504	0.782
	HE	0.049	0.005	90,504	0.824
Technology	University	0.063	0.003	211,145	0.790
	HE	0.047	0.003	211,145	0.860

Target Graduation Effects by Field

RDD: Graduation from Target Program within 8 Years by Field



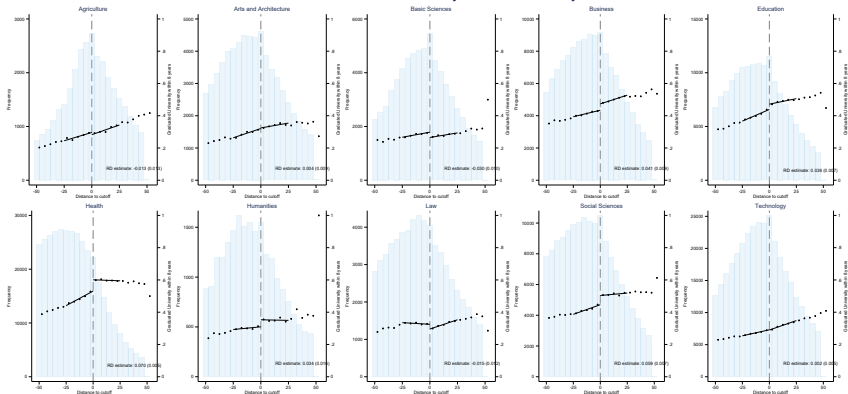
- Crossing the cutoff increases target-program graduation across all fields.
- The largest effects appear in Health, Business, Social Sciences, Education, and Law.

RDD estimates by field: target graduation

Field	Estimate	SE
Agriculture	0.087	0.011
Arts & Architecture	0.123	0.010
Basic Sciences	0.070	0.008
Business	0.224	0.013
Education	0.170	0.007
Health	0.283	0.008
Humanities	0.136	0.017
Law	0.151	0.016
Social Sciences	0.224	0.009
Technology	0.113	0.004

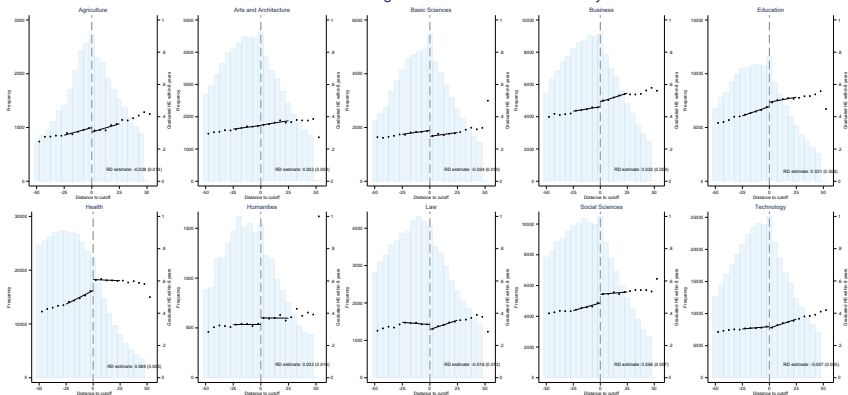
Graduation effects by field: university

RDD: Graduation from University within 8 Years by Field



Graduation effects by field: any higher education

RDD: Graduation from Higher Education within 8 Years by Field



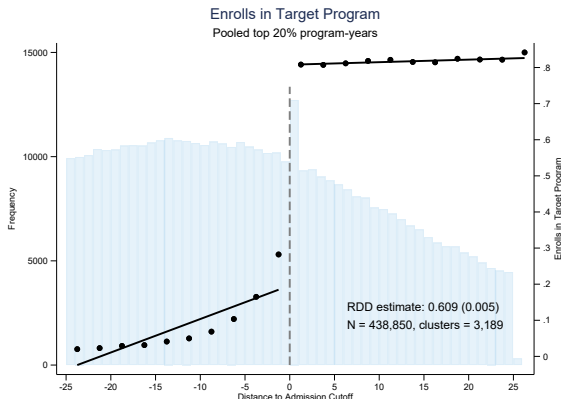
RDD estimates by field: university and HE graduation

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	-0.013	0.013	21,102	0.274
	HE	-0.028	0.014	21,102	0.312
Arts & Architecture	University	0.004	0.009	39,715	0.292
	HE	0.002	0.009	39,715	0.335
Basic Sciences	University	-0.030	0.010	43,849	0.283
	HE	-0.034	0.010	43,849	0.303
Business	University	0.041	0.009	79,964	0.414
	HE	0.032	0.009	79,964	0.448
Education	University	0.036	0.007	95,508	0.403
	HE	0.031	0.006	95,508	0.434
Health	University	0.070	0.005	206,050	0.484
	HE	0.068	0.005	206,050	0.497
Humanities	University	0.034	0.016	13,470	0.301
	HE	0.032	0.016	13,470	0.330
Law	University	-0.015	0.012	37,000	0.330
	HE	-0.016	0.012	37,000	0.337
Social Sciences	University	0.059	0.007	90,504	0.417
	HE	0.056	0.007	90,504	0.440
Technology	University	0.002	0.005	211,145	0.273
	HE	-0.007	0.005	211,145	0.310

RDD: Threshold crossing effects

Selective Program-Years

Selective Programs: Target Enrollment

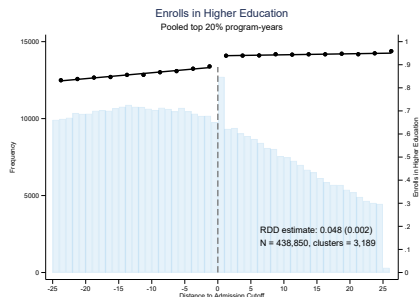
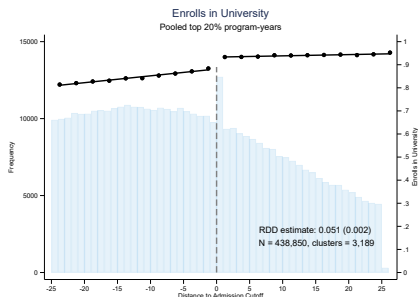


Selective programs

- Selectivity is based on the Language–Math score distribution of admitted students.
- Programs are classified as selective if they appear at least once in the top 20 % of the pooled post-2012 selectivity ranking.

Crossing the cutoff in selective programs increases target-program enrollment by approximately 61 percentage points.

Selective Programs: Broader Enrollment Outcomes



- Effects on university and higher-education enrollment remain much smaller than target-program enrollment.
- Even among selective programs, cutoff crossing mainly changes **where students enroll**, not overall higher-education participation.

RDD: Threshold crossing effects

Selectivity Gaps and Next-Best Alternatives

Construction of Delta Selectivity

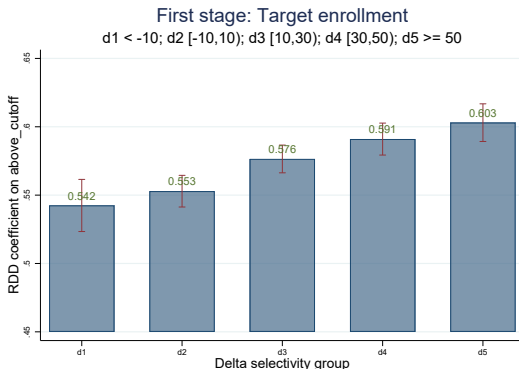
- For each target application, we identify the student's next-best feasible alternative.
- Program selectivity is measured by the average Language–Math score among regularly admitted students.
- We define:

$$\Delta Selectivity = Selectivity_{target} - Selectivity_{nextbest}.$$

- Positive values indicate that the target program is more selective than the next-best alternative.

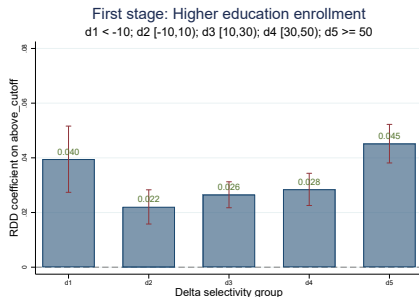
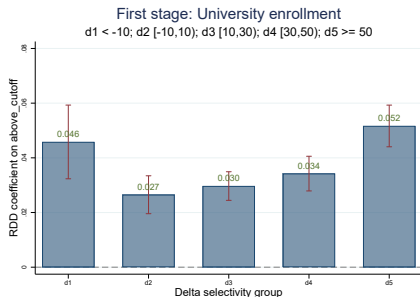
Group	Definition
d1	$\Delta < -10$
d2	$[-10, 10)$
d3	$[10, 30)$
d4	$[30, 50)$
d5	≥ 50

Target Enrollment by Selectivity Gap



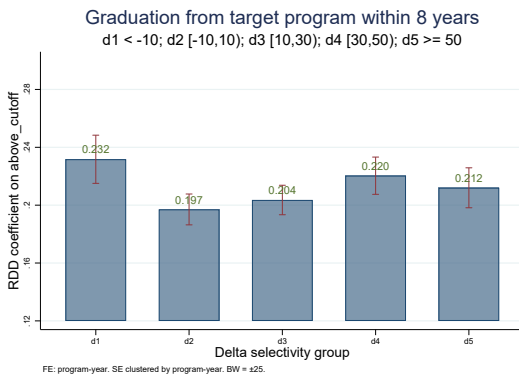
- Crossing the cutoff increases target-program enrollment across all groups.
- Effects are strongest when the target program is substantially more selective than the next-best alternative.
- Admission matters most when students face large differences in program selectivity.

Broader Enrollment Outcomes by Selectivity Gap



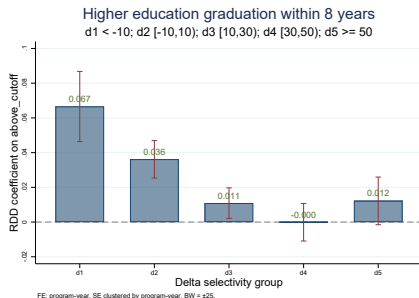
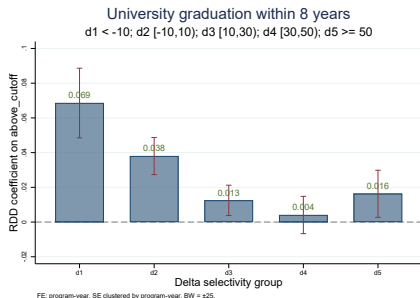
- Effects on university and higher-education enrollment remain relatively small across groups.
- Differences across selectivity-gap groups are much less pronounced than for target-program enrollment.
- The main response margin remains the specific program attended rather than overall higher-education participation.

Target Graduation by Selectivity Gap



- Crossing the cutoff increases target-program graduation across all selectivity-gap groups.
- Target-graduation effects remain large across the distribution of target versus next-best selectivity.

Broader Graduation Outcomes by Selectivity Gap



- Effects on university and higher-education graduation are smaller than target-program graduation effects.
- Broader graduation effects are more heterogeneous across selectivity-gap groups.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

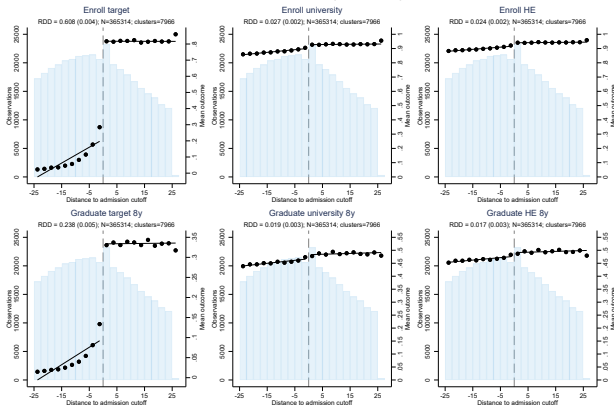
RDD: Threshold crossing effects

Same-field vs Different-field Alternatives

RDD around cutoff: same field

RDD around cutoff: Same field

Binned means, local linear fits, and running-variable histogram. BW = ± 25 .

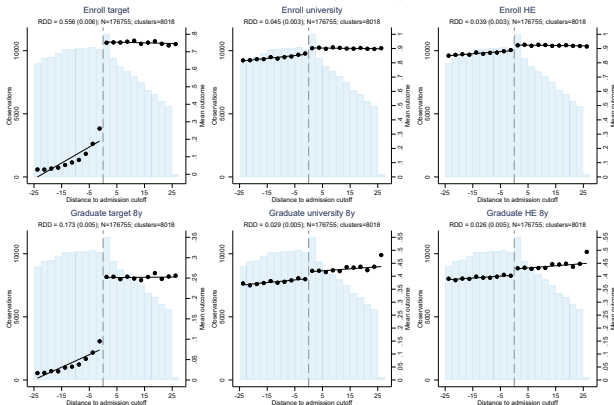


- Crossing the cutoff generates large increases in target-program enrollment and graduation when the next-best alternative belongs to the same field.
- Effects on broader university and higher-education outcomes remain small, suggesting that admission mainly affects program choice within a field.

RDD around cutoff: different field

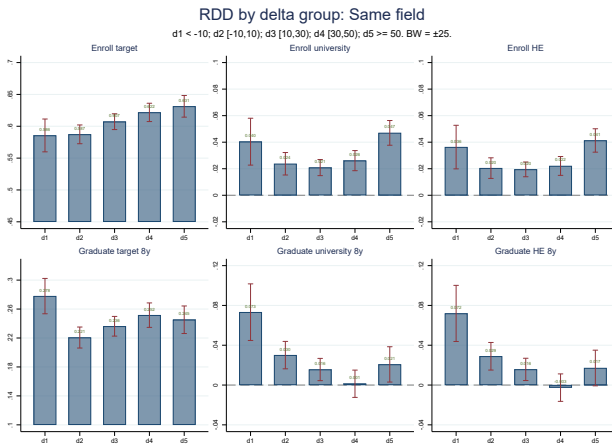
RDD around cutoff: Different field

Binned means, local linear fits, and running-variable histogram. BW = ± 25 .



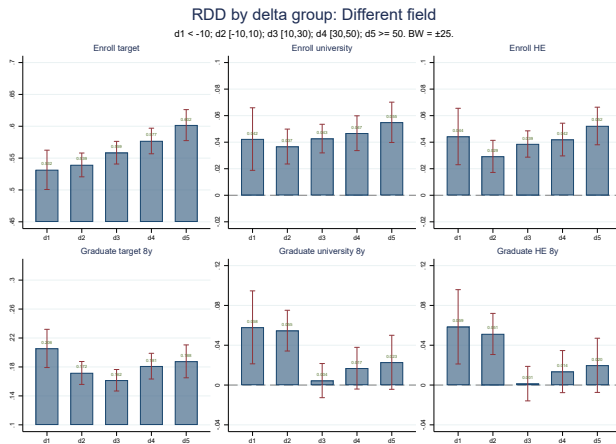
- Target-program enrollment and graduation effects remain large when the next-best alternative belongs to a different field.
- Effects on broader university and higher-education outcomes are somewhat larger than in the same-field sample, consistent with admission affecting both program and field choice.

RDD by selectivity gap: same field



- Within same-field alternatives, target-enrollment effects increase with the selectivity gap between the target and next-best program.
- Broader university and higher-education graduation effects remain comparatively small across groups.

RDD by selectivity gap: different field



- Within different-field alternatives, target-enrollment effects also increase with the selectivity gap.
- Broader graduation effects are less stable across groups, suggesting more heterogeneous substitution patterns when the fallback option is in another field.

Inframarginal Program-Year Analysis

From Marginal to Inframarginal Students

- The previous analysis focused on applicants located around admission cutoffs.
- We now study **inframarginal students**: applicants who would have been admitted to their target program in every year of the relevant period.
- To identify them, we reconstruct each student's application score using the admission weights from each year and compare it with the corresponding admission cutoff.
- The objective is to estimate whether changes in program capacity affect the outcomes of these inframarginal cohorts.

Inframarginal Program-Year Analysis

Inframarginal Design No. 1

Specification

- Following the model notes, program size should be measured using *total enrollment*:

$$\Delta N_{pt}^{total} = \pi \Delta Z_{pt} + \alpha_t + \nu_{pt}.$$

- Outcomes continue to be computed using inframarginal students:

$$\Delta y_{pt}^{infra} = \beta \widehat{\Delta N}_{pt}^{total} + \alpha_t + \varepsilon_{pt}.$$

- N_{pt}^{total} : total enrollment in program p and year t .
- Z_{pt} : total vacancies/cupos.
- y_{pt}^{infra} : graduation rate among inframarginal students.

Corrected Inframarginal IV Results

Period	Instrument	Outcome	First stage	F-stat	Reduced form	2SLS
2007–2012	Δ Total Cupos	Target grad. 8y	0.816 (0.037)	482.57	-0.00038 (0.00024)	-0.00048 (0.00030)
2007–2012	Δ Total Cupos	Univ. grad. 8y	0.816 (0.037)	482.57	0.00063 (0.00030)	0.00079** (0.00035)
2007–2012	Δ Total Cupos	HE grad. 8y	0.816 (0.037)	482.57	0.00063 (0.00030)	0.00079** (0.00035)
2013–2016	Δ Total Cupos	Target grad. 8y	0.543 (0.054)	96.11	-0.00011 (0.00034)	-0.00020 (0.00064)
2013–2016	Δ Total Cupos	Univ. grad. 8y	0.543 (0.054)	96.11	0.00048* (0.00025)	0.00088** (0.00039)
2013–2016	Δ Total Cupos	HE grad. 8y	0.543 (0.054)	96.11	0.00048* (0.00025)	0.00088** (0.00039)

Standard errors in parentheses. First stage: ΔN_{pt}^{total} on ΔZ_{pt} . Reduced form: Δy_{pt}^{infra} on ΔZ_{pt} . 2SLS: Δy_{pt}^{infra} on predicted ΔN_{pt}^{total} . * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

First Stage Robustness and Residual Variation

Panel A. Sequential first-stage specifications

Specification	2007–2012	2013–2016
Differences, no FE	0.817	0.543
Differences + year FE	0.816	0.543
Differences + program FE	0.816	0.637
Differences + program + year FE	0.813	0.638
Levels + program + year FE	0.824	0.525

Panel B. Residual variation after program and year FE

Period	Enrollment SD	Residual enrollment SD	Cupos SD	Residual Cupos SD
2007–2012	45.5	31.8	49.6	33.5
2013–2016	62.4	37.9	67.7	40.0

- Adding fixed effects does not eliminate the first stage.
- Residual variation remains substantial in both periods.
- The weaker 2013–2016 first stage does not appear to be driven by a lack of within-program variation.

Inframarginal Student Definitions

Identify students who would have enrolled in the target program regardless of small changes in annual admission capacity.

Construction

- For each program, compute the annual number of students who actually enrolled.
- Define an enrollment threshold using one of two alternative rules:
 - ▶ **First Cohort:** size of the first observed enrolled cohort.
 - ▶ **Minimum Cohort:** minimum enrolled cohort observed over the sample period.
- An applicant is classified as inframarginal if their admission rank is at or above the threshold defined for that program.

$$\text{Inframarginal}_{ipt} = 1 \iff \text{Rank}_{ipt} \leq T_p,$$

where T_p is either the first enrolled cohort size or the minimum enrolled cohort size for program p .

Results

	First Cohort	80 % First Cohort	Minimum Cohort	80 % Minimum Cohort
Infra students (N)	388,867	314,482	354,631	284,126
Share ranked infra (%)	72.90	58.96	66.48	53.27
FD observations	6,523	6,523	6,521	6,519
Programs	1,257	1,257	1,257	1,257
First stage	0.7317*** (0.0504)	0.7317*** (0.0504)	0.7317*** (0.0504)	0.7317*** (0.0504)
F-stat	210.7	210.7	210.6	210.6
2SLS Program	0.00009 (0.00022)	0.00011 (0.00023)	0.00018 (0.00027)	0.00019 (0.00029)

Notes: Columns differ only in the definition of inframarginal students. In all columns, the threshold used to define inframarginal status is based on effective enrollment. **First Cohort** defines inframarginal students as applicants whose admission rank is weakly below the number of students effectively enrolled in the first observed cohort of each program. **80 % First Cohort** restricts this threshold to 80 percent of the first enrolled cohort. **Minimum Cohort** uses the minimum annual effective enrollment observed for each program over the sample period. **80 % Minimum Cohort** restricts this minimum threshold to 80 percent. The first stage instruments annual changes in total program enrollment with annual changes in total vacancies. The outcome is graduation from the program in which the student initially enrolled within eight years. Robust standard errors clustered at the program level are reported in parentheses.

Alternative inframarginal

Definition	N	Share (%)	FD Obs.	First stage	F-stat	2SLS
<i>First Cohort</i>						
100 %	388,867	70.59	6,523	0.7317*** (0.0504)	210.7	0.00009 (0.00022)
90 %	350,913	63.70	6,523	0.7317*** (0.0504)	210.7	0.00011 (0.00023)
80 %	314,482	57.09	6,523	0.7317*** (0.0504)	210.7	0.00011 (0.00023)
70 %	276,422	50.18	6,521	0.7317*** (0.0504)	210.7	0.00009 (0.00023)
60 %	238,522	43.30	6,519	0.7317*** (0.0504)	210.7	0.00009 (0.00024)
<i>Minimum Cohort</i>						
100 %	354,631	64.37	6,521	0.7317*** (0.0504)	210.6	0.00018 (0.00027)
90 %	318,091	57.74	6,519	0.7317*** (0.0504)	210.6	0.00021 (0.00028)
80 %	284,126	51.58	6,519	0.7317*** (0.0504)	210.6	0.00019 (0.00029)
70 %	249,085	45.21	6,513	0.7317*** (0.0504)	210.6	0.00014 (0.00029)
60 %	214,682	38.97	6,511	0.7317*** (0.0504)	210.6	0.00020 (0.00030)

Notes: The table reports robustness checks using progressively more conservative definitions of inframarginal students. For each program, the baseline definition uses either the size of the first enrolled cohort (*First Cohort*) or the minimum enrolled cohort observed over 2007–2016 (*Minimum Cohort*). Alternative definitions restrict the inframarginal sample to the top 90 %, 80 %, 70 %, and 60 % of each baseline cohort size. All thresholds are based on effective enrollment and then applied to applicants according to their admission ranking. The first stage instruments annual changes in total enrollment with annual changes in total vacancies. The outcome is graduation from the initially enrolled program within eight years. Robust standard errors clustered at the program level are reported in parentheses.

Heterogeneity by University Selectivity: First Cohort

University	Mean PSU	First stage	F-stat	2SLS
<i>More selective universities</i>				
Univ. Católica	685.8	0.9925*** (0.0277)	1,284.7	-0.00023 (0.00036)
Univ. de Chile	675.7	0.8039*** (0.0448)	322.6	-0.00003 (0.00031)
Univ. de Santiago	616.7	0.8508*** (0.0315)	728.1	0.00113** (0.00053)
<i>Less selective universities</i>				
Univ. Playa Ancha	548.4	0.6235*** (0.0545)	131.1	-0.01385*** (0.00282)
Univ. Católica de Concepción	558.0	0.4448*** (0.0516)	74.3	0.00005 (0.00260)
Univ. del Bío-Bío	560.5	0.6681*** (0.1913)	12.2	-0.01598** (0.00797)

Notes: Results use the 2007–2016 panel and the *First Cohort* definition of inframarginal students. Mean PSU is the average Language–Mathematics PSU score of students enrolled in their target program. The first stage estimates the effect of annual changes in total vacancies on annual changes in total program enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student induced by changes in vacancies on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Heterogeneity by University Selectivity: Minimum Cohort

University	Mean PSU	First stage	F-stat	2SLS
<i>More selective universities</i>				
Univ. Católica	685.8	0.9925*** (0.0277)	1,284.7	-0.00023 (0.00037)
Univ. de Chile	675.7	0.8039*** (0.0448)	322.6	-0.00008 (0.00031)
Univ. de Santiago	616.7	0.8508*** (0.0315)	728.1	0.00113** (0.00053)
<i>Less selective universities</i>				
Univ. Playa Ancha	548.4	0.6235*** (0.0545)	131.1	-0.01447*** (0.00293)
Univ. Católica de Concepción	558.0	0.4448*** (0.0516)	74.3	0.00007 (0.00285)
Univ. del Bío-Bío	560.5	0.6681*** (0.1913)	12.2	-0.01756** (0.00837)

Notes: Results use the 2007–2016 panel and the *Minimum Cohort* definition of inframarginal students. Mean PSU is the average Language–Mathematics PSU score of students enrolled in their target program. The first stage estimates the effect of annual changes in total vacancies on annual changes in total program enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student induced by changes in vacancies on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Program selectivity and graduation effects

Panel A. First Cohort

Program selectivity	Programs	FD obs.	First Stage	F-stat	2SLS
PSU < 550	410	1,400	0.4356*** (0.0296)	217.0	0.00125 (0.00112)
PSU 550–600	467	2,563	0.5165*** (0.0369)	196.3	0.00036 (0.00071)
PSU 600–650	249	1,665	0.7510*** (0.0486)	238.9	-0.00032 (0.00053)
PSU ≥ 650	131	895	0.9229*** (0.0326)	802.3	-0.00009 (0.00018)

Panel B. Minimum Cohort

Program selectivity	Programs	FD obs.	First Stage	F-stat	2SLS
PSU < 550	410	1,400	0.4356*** (0.0296)	217.0	0.00140 (0.00117)
PSU 550–600	467	2,561	0.5162*** (0.0369)	196.0	0.00048 (0.00077)
PSU 600–650	249	1,665	0.7510*** (0.0486)	238.9	-0.00023 (0.00052)
PSU ≥ 650	131	895	0.9229*** (0.0326)	802.3	-0.00004 (0.00025)

Notes: Programs are classified according to the average Language–Mathematics PSU score of enrolled students over 2007–2016: PSU < 550, 550–600, 600–650, and PSU ≥ 650. Standard errors clustered at the program level are reported in parentheses. Outcomes correspond to graduation from the initially enrolled program within eight years.

Heterogeneity by Field of Study

Field group	Programs	FD obs.	First stage	F-stat	2SLS
Panel A. First Cohort					
Business and Law	130	651	0.6305*** (0.0654)	93.0	0.00179** (0.00084)
Health	221	1,290	0.4074*** (0.0342)	142.1	-0.00013 (0.00082)
STEM	432	2,162	0.8502*** (0.0475)	320.7	-0.00015 (0.00010)
Education and Humanities	372	1,913	0.7158*** (0.0994)	51.9	-0.00060 (0.00061)
Arts, Architecture and Agriculture	102	507	0.6541*** (0.0557)	138.0	0.00070 (0.00120)
Panel B. Minimum Cohort					
Business and Law	130	651	0.6305*** (0.0654)	93.0	0.00189** (0.00084)
Health	221	1,288	0.4069*** (0.0342)	141.9	0.00013 (0.00084)
STEM	432	2,162	0.8502*** (0.0475)	320.7	-0.00014 (0.00012)
Education and Humanities	372	1,913	0.7158*** (0.0994)	51.9	-0.00062 (0.00064)
Arts, Architecture and Agriculture	102	507	0.6541*** (0.0557)	138.0	0.00073 (0.00117)

Notes: Results use the 2007–2016 panel. Fields are grouped using the target program's field of study and fixed at the program level. The first stage estimates the effect of annual changes in vacancies on annual changes in total enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student, induced by changes in vacancies, on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Inframarginal Students: Graduation at 8 vs. 10 Years

Definition	Horizon	Programs	FD obs.	First stage	2SLS
First Cohort	8 years	1,092	4,881	0.7413*** (0.0579)	0.00002 (0.00021)
First Cohort	10 years	1,092	4,881	0.7413*** (0.0579)	0.00031 (0.00034)
Minimum Cohort	8 years	1,092	4,881	0.7413*** (0.0579)	0.00010 (0.00027)
Minimum Cohort	10 years	1,092	4,881	0.7413*** (0.0579)	0.00048 (0.00043)

Notes: The table compares graduation outcomes measured within eight and ten years using the same 2007–2014 sample. The outcome is graduation from the program in which the student initially enrolled. Each estimate uses 1,092 programs and 4,881 first-difference observations. The first stage instruments annual changes in total program enrollment with annual changes in total vacancies; the first-stage coefficient implies that 10 additional vacancies increase enrollment by approximately 7.4 students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Inframarginal IV results

Outcome	First stage	F-stat	Reduced form	2SLS
Target graduation, 8y	0.732 (0.050)	210.7	0.00013 (0.00020)	0.00019 (0.00028)
University graduation, 8y	0.732 (0.050)	210.7	0.00005 (0.00012)	0.00007 (0.00016)
Higher education graduation, 8y	0.732 (0.050)	210.7	0.00005 (0.00012)	0.00007 (0.00016)

- 494,675 inframarginal students (90.6 % of enrollment).

Heterogeneity by university selectivity

University	Mean PSU	First stage	F-stat	Reduced form	2SLS
<i>Highly selective universities</i>					
UC	693.1	0.993 (0.028)	903.9	0.00024 (0.00028)	0.00024 (0.00028)
UCH	686.1	0.804 (0.045)	390.5	0.00096*** (0.00026)	0.00119*** (0.00036)
<i>Less selective universities</i>					
UNAB	559.2	0.380 (0.040)	90.8	-0.00029 (0.00022)	-0.00076 (0.00060)
UCSC	563.5	0.445 (0.052)	75.4	-0.00174** (0.00076)	-0.00385** (0.00157)
UBB	567.5	0.668 (0.191)	14.1	-0.00464** (0.00206)	-0.00673* (0.00366)

- Mean PSU is based on target-program selectivity.
- Outcome: university graduation among inframarginal students.
- Negative effects emerge mainly among less selective institutions.

Inframarginal Program-Year Analysis

Inframarginal Design No. 2

Entry of private universities into the SUA

Research question

Did first-year enrollment in incumbent CRUCH programs change after eight private universities entered the SUA in 2012, particularly in programs with greater pre-existing exposure to these entrants?

Sample

- Incumbent CRUCH program $\times$ admission-year panel, 2007–2016.
- Outcome: number of students entering the first year of program p in year t .

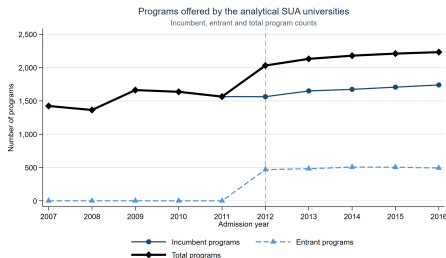
Exposure design

- Markets are defined by pre-treatment region $\times$ academic field.
- Exposure measures the 2009–2011 enrollment presence of the future entrant universities in the incumbent program's market.
- Total exposure uses the entrant enrollment share; Triangular and Gaussian measures additionally weight entrants by PSU-score proximity.
- Exposure is fixed over time and interacted with the post-entry period, 2012–2016.

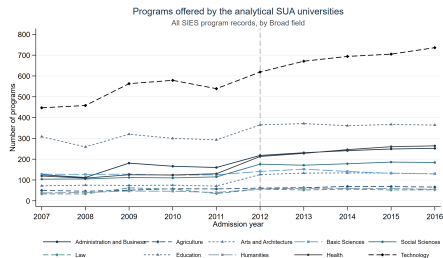
The design compares pre/post-2012 enrollment changes across incumbent programs with different levels of exposure to the future entrants.

Evolution of the number of SUA programs

Programs by university status



Programs by Broad field



- The incorporation of the eight entrant universities adds roughly 470 programs in 2012. Their program supply remains relatively stable afterward, while incumbents account for most subsequent growth.
- Technology represents the largest field and an important share of program expansion.

Construction of the exposure measures

1. Pre-reform enrollment in market m

$$A_m = \sum_{\substack{q \in \mathcal{E} \\ m(q)=m}} \bar{N}_{q,2009-2011}^{\text{first}}, \quad T_m = \sum_{\substack{q \\ m(q)=m}} \bar{N}_{q,2009-2011}^{\text{first}}$$

- $\mathcal{E}$ is the set of entrant programs and $m(q)$ is program q 's pre-treatment region $\times$ field market.
- $\bar{N}_{q,2009-2011}^{\text{first}}$ is average first-year enrollment: A_m is entrant and T_m total enrollment in m .

3. Selectivity-weighted exposure

$$E_p^k = \frac{\sum_{\substack{q \in \mathcal{E} \\ m(q)=m(p)}} k(|S_p - S_q|) \bar{N}_{q,2009-2011}^{\text{first}}}{T_{m(p)}}$$

- S_p, S_q : pre-reform mean PSU scores of incumbent p and entrant q ; $|S_p - S_q|$ is their selectivity distance.
- $k \in \{\text{Triangular, Gaussian}\}$ gives greater weight to closer entrants, so E_p^k varies within a market.

2. Total exposure

$$E_p = \frac{A_{m(p)}}{T_{m(p)}}$$

- Entrant share of pre-reform first-year enrollment in the market of incumbent program p .
- Incumbents in the same region $\times$ field market share the same E_p .

4. Kernel-weighted denominator

$$E_p^{k,\text{KD}} = \frac{\sum_{\substack{q \in \mathcal{E} \\ m(q)=m(p)}} k(|S_p - S_q|) \bar{N}_{q,2009-2011}^{\text{first}}}{\sum_{\substack{\ell \\ m(\ell)=m(p)}} k(|S_p - S_\ell|) \bar{N}_{\ell,2009-2011}^{\text{first}}}$$

- Same numerator as in 3; the denominator weights every program ℓ in the market (incumbents, including p , and entrants) by its selectivity distance to p .
- $E_p^{k,\text{KD}} \in [0, 1]$ is the entrant share of the enrollment that is close to p in selectivity.

SUA exposure: samples and descriptive statistics

Panel A: Analytical samples

Variable	Broad	ISCED-97	Generic
Programs	949	949	945
Program-year observations	8,041	8,030	8,002
Incumbent universities	25	25	25
Pre-treatment markets	130	215	638
Mean annual enrollment, 2009–2011	53.2	53.2	53.2
Mean PSU score, 2009–2011	588.8	588.8	589.0

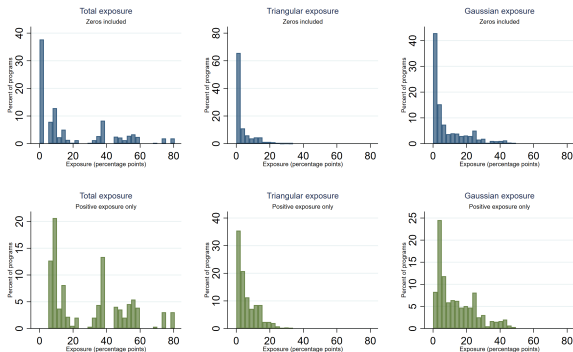
Panel B. Exposure by academic-field definition

Field definition	Exposure measure	Mean	Std. dev.	Share zero
Broad	Total	18.53	22.21	37.7 %
	Triangular	3.48	5.57	46.8 %
	Gaussian	8.58	11.26	37.7 %
ISCED-97	Total	17.88	22.96	43.0 %
	Triangular	3.16	5.79	56.9 %
	Gaussian	8.20	11.70	43.0 %
Generic	Total	13.27	23.40	66.2 %
	Triangular	1.83	6.13	83.4 %
	Gaussian	5.83	11.90	67.4 %

Notes: Panel A reports the analytical sample under each field definition. Enrollment is the unweighted mean across programs of average annual first-year enrollment in 2009–2011; PSU is the program's mean score over the same period. Panel B reports unweighted exposure moments in percentage points, including zero-exposure programs. Total is the entrant share of market enrollment; Triangular and Gaussian additionally weight entrant enrollment by PSU proximity using a 50-point bandwidth. Markets combine pre-treatment region with the corresponding field definition.

SUA exposure distributions: Broad field

One observation per incumbent program



Top row includes zeros; bottom row conditions on positive exposure.

- Each incumbent program appears once. The horizontal axis reports exposure in percentage points and the vertical axis the share of programs.
- The upper row includes all programs; the lower row excludes zeros and shows the distribution conditional on positive exposure.
- Total exposure has the largest values. Triangular weighting is the most restrictive, while Gaussian weighting produces an intermediate distribution.

First-stage specification

$$N_{pt}^{\text{first}} = \pi_k (E_p^k \times \text{Post}_t) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

- p indexes incumbent programs and t years. N_{pt}^{first} is first-year enrollment of program p in year t , while E_p^k is its pre-reform exposure and $\text{Post}_t = 1$ in 2012–2016.
- $f(p)$ denotes the pre-treatment field of program p and $r(p)$ its pre-treatment region. Accordingly, $\alpha_{f(p),t}$ are field $\times$ year fixed effects and $\gamma_{r(p),t}$ are region $\times$ year fixed effects.
- μ_p denotes program fixed effects, and π_k is the post-2012 enrollment difference associated with a 10 percentage-point increase in exposure measure k .

Exposure is positive in only four regions

- In the 2009–2011 analytical sample, entrant universities offered regular undergraduate programs only in Metropolitana, Valparaíso, Biobío and La Araucanía.
- Thus, these are the only regions containing entrant programs and positive entrant enrollment before integration into the SUA.
- In every other region, the entrant-enrollment numerator is zero: $A_m = 0$, and therefore $E_p = A_{m(p)} / T_{m(p)} = 0$.

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times \text{Post}_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	1.967*** (0.336)	1.893*** (0.335)	1.872*** (0.289)	0.798 (0.662)	0.904* (0.510)	0.324 (0.338)
Wald <i>F</i>	34.36	31.92	41.90	1.45	3.14	0.92
<i>Triangular</i>	0.935 (0.972)	1.392 (0.934)	2.185** (0.927)	-5.252** (2.115)	-3.415*** (1.310)	-0.975 (0.829)
Wald <i>F</i>	0.93	2.22	5.56	6.17	6.80	1.39
<i>Gaussian</i>	1.775*** (0.439)	1.939*** (0.450)	2.342*** (0.533)	-2.767* (1.401)	-1.447* (0.832)	-0.558 (0.571)
Wald <i>F</i>	16.36	18.60	19.33	3.90	3.03	0.96
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: The dependent variable is first-year enrollment in incumbent CRUCH programs. Total exposure is entrants' pre-treatment enrollment share; Triangular and Gaussian additionally weight entrant enrollment by PSU-score proximity using a 50-point bandwidth. Markets combine pre-treatment region with the indicated field definition. Exposure is interacted with Post_t , equal to one in 2012–2016, and measured in 10 percentage-point units. All regressions absorb program and corresponding field × year fixed effects; the last three columns additionally absorb region × year fixed effects. Standard errors are clustered by the corresponding pre-treatment market and reported in parentheses. Wald *F* statistics test the exposure coefficient. Sample differences reflect singleton removal by `reghdfe`. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$. Robustness: [regions with entrant universities only](#) → [positive exposure only](#) → [selectivity trends](#) →

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^k} \times \text{Post}_t \right] + \theta_k \left[D_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log E_p^k} = \begin{cases} \log(E_p^k), & E_p^k > 0, \\ 0, & E_p^k = 0, \end{cases} \quad D_p^k = \mathbf{1}\{E_p^k > 0\}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.100*** (0.034)	0.072** (0.035)	0.021 (0.030)	−0.005 (0.039)	−0.024 (0.032)	−0.022 (0.027)
<i>F</i>	8.84	4.24	0.48	0.01	0.55	0.69
<i>Triangular</i>	−0.013 (0.015)	0.004 (0.016)	0.016 (0.021)	−0.050*** (0.017)	−0.024 (0.016)	−0.005 (0.017)
<i>F</i>	0.81	0.07	0.59	8.95	2.37	0.07
<i>Gaussian</i>	−0.026 (0.032)	−0.024 (0.030)	0.001 (0.018)	−0.107*** (0.030)	−0.079*** (0.026)	−0.025 (0.017)
<i>F</i>	0.68	0.64	0.01	12.72	9.35	2.19
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921

Notes: Each coefficient reports β_k , the post-2012 elasticity with respect to exposure intensity. Zero-exposure programs are retained by setting $\widetilde{\log E_p^k} = 0$ when $E_p^k = 0$ and including $D_p^k \times \text{Post}_t$ separately; the coefficient on this indicator is not reported. Thus, β_k captures variation in exposure intensity among positively exposed programs while allowing zero-exposure programs to remain in the estimation sample. All specifications absorb program and corresponding field × year fixed effects; the last three columns additionally absorb pre-treatment region × year fixed effects. Standard errors are clustered at the pre-treatment market level. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$. Robustness: regions with entrant universities only → positive exposure only → selectivity trends →

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k, \text{KD}} \times \text{Post}_t \right) + \mu_p + \alpha_{f_m(p), t} + \gamma_{r(p), t} + \nu_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Triangular KD</i>	0.659** (0.299)	0.717** (0.292)	1.371*** (0.481)	-1.726** (0.737)	-1.286*** (0.427)	-0.208 (0.470)
Wald F	4.86	6.03	8.13	5.48	9.08	0.20
Programs with zero exposure	444	540	767	443	539	766
<i>Gaussian KD</i>	1.326*** (0.346)	1.420*** (0.350)	1.913*** (0.394)	-1.919* (1.051)	-1.049* (0.600)	-0.218 (0.472)
Wald F	14.70	16.42	23.56	3.33	3.06	0.21
Programs with zero exposure	358	408	616	357	407	615
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Exposure uses the kernel-weighted denominator (point 4 of the exposure construction): the denominator weights every program in the market by its selectivity distance to p , with the same 50-point Triangular and Gaussian kernels as the numerator. Sample and fixed effects as in the main first stage; SE clustered by pre-treatment market. Exposure in 10 percentage-point units. As in the [main first stage](#), coefficients are positive with field × year FE and negative with region × year FE. Magnitudes are smaller because KD exposure takes larger values (Broad, 2011 mean: 0.11 for Triangular KD vs. 0.03 for Triangular). Robustness to selectivity trends: [appendix →](#). *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^{k,\text{KD}}} \times \text{Post}_t \right] + \theta_k \left[D_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Triangular KD</i>	-0.001 (0.014)	0.014 (0.018)	0.022 (0.023)	-0.052*** (0.018)	-0.032* (0.017)	-0.014 (0.018)
Wald <i>F</i>	0.00	0.60	0.96	8.86	3.83	0.60
Programs with zero exposure	444	540	767	443	539	766
<i>Gaussian KD</i>	-0.012 (0.036)	-0.018 (0.032)	0.009 (0.019)	-0.138*** (0.037)	-0.095*** (0.028)	-0.030* (0.018)
Wald <i>F</i>	0.12	0.32	0.24	13.59	11.31	3.01
Programs with zero exposure	358	408	616	357	407	615
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Exposure uses the kernel-weighted denominator (point 4 of the exposure construction): the denominator weights every program in the market by its selectivity distance to p , with the same 50-point Triangular and Gaussian kernels as the numerator. Sample and fixed effects as in the main first stage; SE clustered by pre-treatment market. $\widetilde{\log E} = \log E$ if $E > 0$ and 0 otherwise; $D_p^k = \mathbf{1}\{E_p^{k,\text{KD}} > 0\}$. Observations with zero first-year enrollment are excluded. Compare with the [main log first stage](#). *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

First-stage heterogeneity by incumbent selectivity

	PSU < 550		PSU 550–649		PSU $\geq$ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
<i>SUA exposure</i>						
Coefficient	−1,61 (2,78)	−0,54 (2,10)	−3,17** (1,54)	−0,98 (0,83)	−15,75* (8,25)	−5,01 (4,38)
F-statistic	0.34	0.07	4.26	1.41	3.65	1.31
Observations	2,215		4,609		1,188	
Programs	272		531		143	
Clusters	89		100		25	

Notes: Standard errors clustered by pre-treatment market in parentheses. All regressions include program, Broad-field $\times$ year, and region $\times$ year FE. Coefficients correspond to a 10 percentage-point increase in exposure. The same split for conditional similarity is in the [appendix](#). * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Cosine exposure

Each program is represented by a vector describing the distribution of its 2011 first-year students. Each coordinate counts students with a particular PSU profile and, depending on the definition, campus region and Broad field. Programs are more similar when their students are distributed across the same vector cells.

1. Program vectors

$$\mathbf{v}_p^{PSU} = (v_{p,b})_{b \in \mathcal{B}}, \quad \mathbf{v}_p^{PSU \times R} = (v_{p,rb})_{\substack{r \in \mathcal{R}, \\ b \in \mathcal{B}}}, \quad \mathbf{v}_p^{PSU \times R \times F} = (v_{p,rfb})_{\substack{r \in \mathcal{R}, f \in \mathcal{F}, \\ b \in \mathcal{B}}}.$$

Here, b denotes a 25-point PSU bin, r campus region, and f Broad field. Each vector cell contains the number of first-year students in that combination. Entrant-program vectors are constructed analogously.

2. Similarity between incumbent p and entrant j

$$\cos \theta_{j,p}^d = \frac{\mathbf{v}_j^d \cdot \mathbf{v}_p^d}{\|\mathbf{v}_j^d\| \|\mathbf{v}_p^d\|}, \quad d \in \{PSU, PSU \times R, PSU \times R \times F\}.$$

The cosine compares the **shape of the student distributions**, rather than program size. Under interaction-cell definitions, similarity is zero when programs do not share the indicated region or field.

3. Additive similarity measures

$$\cos \theta_{j,p}^{PSU+R} = \frac{\cos \theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\}}{2}, \quad \cos \theta_{j,p}^{PSU+R+F} = \frac{\cos \theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\} + \mathbf{1}\{f_j = f_p\}}{3}.$$

These measures give equal weight to PSU-profile similarity and coincidence in campus region and, where applicable, Broad field.

4. Exposure of incumbent program p

$$E_p^d = \sum_{j \in \mathcal{J}} w_j \cos \theta_{j,p}^d, \quad w_j = \frac{N_{j,2011}^{\text{first}}}{\sum_{\ell \in \mathcal{J}} N_{\ell,2011}^{\text{first}}}.$$

Exposure is higher when incumbent p resembles entrant programs that account for a larger share of entrant enrollment.

Notes: $\mathcal{J}$ contains programs from the eight entrant universities. Campus region is the university program's location in 2011. Student school region and SES are not used.

Cosine exposure: samples and descriptive statistics

Panel A: Analytical samples

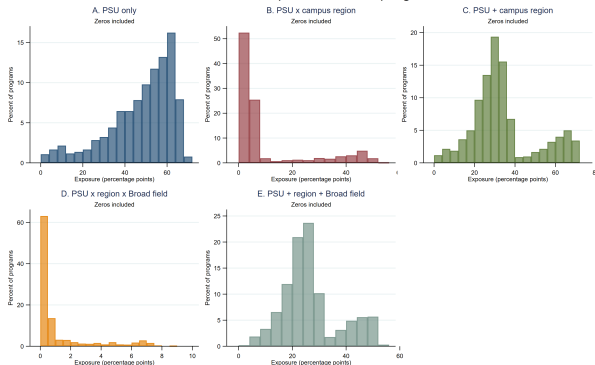
Variable	Full sample	$N_{p,2011}^{\text{first}} \geq 10$
Programs	996	939
Program-year observations	9,105	8,618
Incumbent universities	25	25
Total first-year enrollment, 2011	54,659	54,303
Mean enrollment per program, 2011	54.9	57.8
Mean annual enrollment, 2007–2011	54.8	57.3
Mean PSU score, 2011	578.0	580.9

Panel B: Distribution of raw cosine exposure

Exposure measure	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
	Mean	Std. dev.	Share zero	Mean	Std. dev.	Share zero
PSU bins	0.4762	0.1559	0.0 %	0.4842	0.1540	0.0 %
PSU bins $\times$ campus region	0.0977	0.1495	37.0 %	0.1024	0.1521	34.7 %
PSU bins + campus region	0.3405	0.1614	0.0 %	0.3490	0.1604	0.0 %
PSU bins $\times$ campus region $\times$ Broad field	0.0113	0.0196	38.3 %	0.0119	0.0200	35.8 %
PSU bins + campus region + Broad field	0.2681	0.1078	0.0 %	0.2745	0.1064	0.0 %

Notes: The unit is an incumbent SUA program and exposure moments are unweighted across programs. The restricted sample excludes 57 programs with fewer than 10 first-year students in 2011. Multiplicative measures require coincidence in the indicated dimensions; additive measures assign equal weight to PSU similarity, campus-region coincidence and, where applicable, Broad-field coincidence. The restriction is applied after exposure construction, so entrant profiles, weights, and exposure scaling remain unchanged. Campus region is the university program's location; student school region and SES are not used.

Cosine exposure distributions One observation per incumbent program



Zeros included. Exposure measured in percentage points; each panel uses its own x-axis scale.

- Each incumbent program appears once. The horizontal axis reports cosine exposure in percentage points and the vertical axis the share of programs.
- Multiplicative definitions are more restrictive. In particular, $\text{PSU} \times \text{campus region} \times \text{Broad field}$ places substantial mass at zero and has a much smaller support than the PSU-only and additive measures.

$$N_{pt}^{\text{first}} = \beta_k(E_p^k \times \text{Post}_t) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

Exposure measure	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
	(1) + year	(2) + field × year	(3) + region × year	(4) + year	(5) + field × year	(6) + region × year
PSU bins	−1,811* (0,948)	−1,403 (1,018)	−0,716 (1,002)	−2,116** (1,017)	−1,597 (1,098)	−0,630 (1,075)
Wald F	3,65	1,90	0,51	4,33	2,11	0,34
PSU bins × region	3,976*** (0,601)	4,126*** (0,598)	−5,174 (3,736)	3,903*** (0,611)	4,119*** (0,611)	−5,267 (3,797)
Wald F	43,82	47,55	1,92	40,82	45,41	1,92
PSU bins × region × field	3,680*** (0,658)	3,624*** (0,647)	−3,922* (2,075)	3,590*** (0,666)	3,612*** (0,655)	−4,023* (2,140)
Wald F	31,33	31,36	3,57	29,09	30,44	3,53
PSU bins + region	4,299*** (0,643)	4,416*** (0,641)	−1,476 (2,066)	4,344*** (0,667)	4,577*** (0,668)	−1,298 (2,217)
Wald F	44,76	47,54	0,51	42,36	47,02	0,34
PSU bins + region + field	4,639*** (0,656)	4,429*** (0,642)	−1,480 (2,071)	4,728*** (0,690)	4,590*** (0,669)	−1,302 (2,224)
Wald F	50,07	47,54	0,51	46,97	47,02	0,34
Observations	9,105	9,105	9,103	8,618	8,618	8,610
Programs	996	996	996	939	939	938

Notes: The dependent variable is incumbent-program first-year enrollment. Coefficients are reported first, program-clustered standard errors are in parentheses, and Wald F statistics appear in the corresponding labeled row. Exposure is measured in SD units and $\text{Post}_t = 1$ in 2012–2016. Columns (1) and (4) include program and year FE; Columns (2) and (5) use Broad-field × year FE; Columns (3) and (6) add campus-region × year FE. Multiplicative measures require coincidence in the indicated dimensions; additive measures combine them with equal weights. Columns (4)–(6) exclude programs with fewer than 10 first-year students in 2011. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

Inframarginal Program-Year Analysis

Inframarginal Design No. 3

Design III: sudden vacancy expansions at competing programs

Research question

Does first-year enrollment in an incumbent program k change when competing programs j suddenly expand their regular-admission vacancies, and more so when those programs are closer competitors of k ?

Sample

- Incumbent program $\times$ admission-year panel, 2007–2016 (programs with 2007–2009 entrants).
- Outcome: first-year enrollment N_{kt}^{first} (Formulario D).

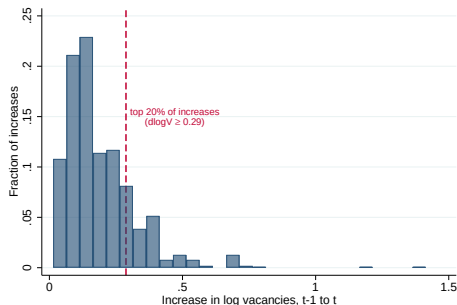
Shock at program j

- Regular vacancies V_{jt} from DEMRE Oferta Académica.
- $\Delta \log V_{jt}$ in the top 20 % of positive changes (pooled 2008–2016) and *sudden*: j existed two years before, was not a top-20 % change in $t - 1$, and the increase is not reversed in $t + 1$.
- Shocks dated 2010–2016; SUA entrants are never shock sources. 179 shocks.

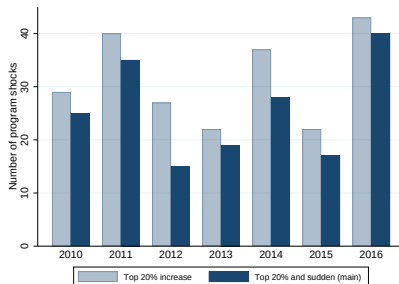
Exposure design

- Same similarity measures as Design II: region $\times$ field markets (Broad, ISCED-97, Generic), Total, Triangular and Gaussian PSU kernels, and cosine similarity of PSU distributions.
- Unlike Design II, exposure varies over time: it accumulates the seats added by competitor shocks since 2010.

Design III: distribution of vacancy changes



Program-years 2010-2016 with vacancies in t-1 and t: N = 6876. 71% have no change, 15% increase (1008, shown; trimmed at 1.5). Threshold: 80th percentile of positive changes, pooled 2008-2016.



- 71 % of program-years have no change in regular vacancies and 15 % increase. The top-20 % threshold is $\Delta \log V \geq 0.29 = \log(4/3)$: vacancies are round numbers, so many increases are exactly +33 %.
- The sudden conditions remove about a fifth of top-20 % increases. Shocks are spread over 2010–2016.

Design III: where are the shocks?

	Program-years	Shocks	Shock rate (%)	Share of shocks (%)
<i>Pre-period selectivity (entrant PSU quintile)</i>				
No PSU data	809	20	2.5	11.2
Q1 (least selective)	976	31	3.2	17.3
Q2	1,150	26	2.3	14.5
Q3	1,288	29	2.3	16.2
Q4	1,331	40	3.0	22.3
Q5 (most selective)	1,322	33	2.5	18.4
<i>Field of study</i>				
Tecnología	2,015	55	2.7	30.7
Educación	1,441	36	2.5	20.1
Administración y Comercio	509	19	3.7	10.6
Salud	886	19	2.1	10.6
Ciencias Básicas	565	14	2.5	7.8
Arte y Arquitectura	347	13	3.7	7.3
Ciencias Sociales	625	12	1.9	6.7
Humanidades	157	10	6.4	5.6
Derecho	142	1	0.7	0.6
Agropecuaria	189	0	0.0	0.0
<i>University</i>				
Private (CRUCH)	2,563	59	2.3	33.0
State (CRUCH)	4,313	120	2.8	67.0
<i>Year</i>				
2010	929	25	2.7	14.0
2011	936	35	3.7	19.6
2012	946	15	1.6	8.4
2013	980	19	1.9	10.6
2014	1,026	28	2.7	15.6
2015	999	17	1.7	9.5
2016	1,060	40	3.8	22.3
Total	6,876	179	2.6	100.0

Shocks per program, 2010–2016	Programs
0	1,196
1	149
2	15
Total	1,360

- 88 % of programs are never shocked; shocks are almost always one-off.
- Shocks are spread across the selectivity distribution.
- More frequent at state universities and in Humanities; almost absent in Law and Agriculture.

Design III: construction of the exposure measures

1. Seats added by a shock at j and pre-period market size

$$\text{seats}_{jt} = (V_{jt} - V_{j,t-1}) \text{shock}_{jt}, \quad T_m = \sum_{\ell: m(\ell)=m} \bar{N}_{\ell,2007-2009}^{\text{first}}$$

$m(\ell)$ is program ℓ 's campus region $\times$ field market (Broad, ISCED-97 or Generic field).

2. Exposure of incumbent k in year t

$$E_{kt}^w = 100 \times \frac{\sum_{j \in m(k), j \neq k} w(k, j) \text{seats}_{jt}}{T_{m(k)}}, \quad \text{cum} E_{kt}^w = \sum_{s=2010}^t E_{ks}^w.$$

3. Weights, as in Design II (S = mean entrant PSU, 2007–2009; $h = 50$)

$$w^{\text{Total}} = 1, \quad w^{\text{Tri}} = \max\left(0, 1 - \frac{|S_k - S_j|}{h}\right), \quad w^{\text{Gau}} = \exp\left[-\frac{1}{2} \left(\frac{|S_k - S_j|}{h}\right)^2\right].$$

- Total exposure: competitor seats added in k 's market, in percentage points of the market's pre-period enrollment.
- Kernels give more weight to shocks at programs with similar selectivity, so exposure varies within a market.
- $\text{cum} E_{kt}^w$ is the time-varying analogue of $E_p \times \text{Post}_t$ in Design II. For programs created after 2009, S_j uses their first three cohorts.

Design III: samples and descriptive statistics

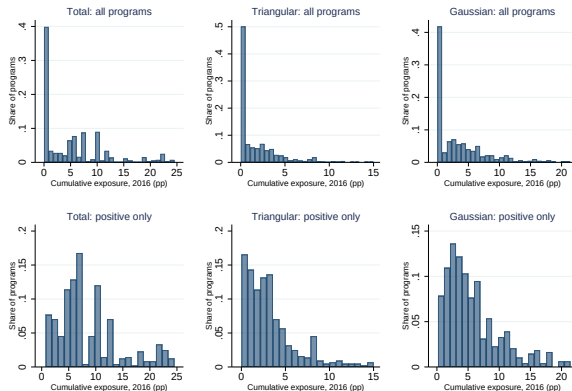
Panel A: Analytical samples

Variable	Broad	ISCED-97	Generic
Programs	989	989	989
Program-year observations	8,925	8,925	8,925
Incumbent universities	25	25	25
Markets (field $\times$ region)	134	221	663
Markets with a shock, 2010–2016	46	61	70
Programs exposed to a market shock	592	501	147
Mean annual enrollment, 2007–2009	51.7	51.7	51.7
Mean PSU score, 2007–2009	583.9	583.9	583.9

Panel B: Cumulative exposure in 2016, by field definition

Field definition	Exposure measure	Mean	Std. dev.	p90	Share zero
Broad	Total	5.79	12.27	13.10	38.2 %
	Triangular	2.00	3.31	5.32	43.6 %
	Gaussian	3.90	6.61	10.46	38.2 %
ISCED-97	Total	5.25	7.50	16.67	47.0 %
	Triangular	2.06	4.03	6.32	53.0 %
	Gaussian	3.73	5.90	11.26	47.0 %
Generic	Total	2.88	8.45	11.90	82.7 %
	Triangular	1.02	4.83	1.53	86.8 %
	Gaussian	2.07	6.84	6.99	82.7 %

Notes: Incumbent programs with 2007–2009 entrants and PSU data, excluding SUA entrants. Panel B reports unweighted moments across programs of cumulative exposure in 2016, in percentage points of pre-period market enrollment, including zero-exposure programs. Kernels use a 50-point PSU bandwidth.



Incumbent programs, cumulative exposure in 2016; trimmed at p99. Broad-field markets.

- Each incumbent program appears once, at its cumulative exposure in 2016 (Broad-field markets).
- The upper row includes all programs; the lower row keeps positive exposure only.
- About 38% of programs are never exposed. Triangular weighting is the most restrictive.

Design III: models to estimate

(1) **TWFE, levels** (mirror of the Design II first stage)

$$N_{kt}^{\text{first}} = \pi_w \text{cum}E_{kt}^w + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \nu_{kt}$$

(2) **TWFE, logs**, zero exposure kept with an indicator

$$\log N_{kt}^{\text{first}} = \beta_w \log \text{cum}E_{kt}^w + \theta_w \mathbf{1}\{\text{cum}E_{kt}^w > 0\} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}$$

(3) **Conditional similarity**, markets with at least one shock, using $\text{cum}E_{kt}^w = M_{kt} \times Q_{kt}^w$

$$N_{kt}^{\text{first}} = \beta_w Q_{kt}^w + \lambda \mathbf{1}\{M_{kt} > 0\} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}$$

(4) **Cosine exposure**: model (1) with $\text{cum}E_{kt}^d$,

$d \in \{PSU, PSU \times R, PSU \times R \times F, PSU + R, PSU + R + F\}$.

(5) **Event studies** around g_k , the first year k is hit (market shock; close-competitor shock; top-5 cosine shock)

$$y_{kt} = \sum_{\ell \neq -1} b_{\ell} \mathbf{1}\{t - g_k = \ell\} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + e_{kt}$$

- Callaway & Sant'Anna with two control groups: never treated only, and not-yet-treated plus never treated. Balanced panel; no field- or region-year effects.
- Own_{kt} : k 's own vacancy shocks to date. Market measures in 10 pp units, SE clustered by market; cosine in SD units, clustered by program.

$$N_{kt}^{\text{first}} = \pi_w \text{cum} E_{kt}^{w,m} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f_m(k),t} + \gamma_{r(k),t} + \nu_{kt}$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.588 (0.677)	2.413** (0.945)	1.109 (0.874)	0.368 (0.362)	1.499* (0.786)	-0.286 (0.815)
Wald <i>F</i>	0.76	6.51	1.61	1.03	3.64	0.12
<i>Triangular</i>	1.572 (1.814)	2.212 (1.434)	0.247 (0.908)	0.544 (1.551)	1.164 (1.046)	-1.594 (1.016)
Wald <i>F</i>	0.75	2.38	0.07	0.12	1.24	2.46
<i>Gaussian</i>	1.105 (1.031)	2.528** (1.021)	1.064 (0.804)	0.673 (0.654)	1.536** (0.759)	-0.545 (0.816)
Wald <i>F</i>	1.15	6.14	1.75	1.06	4.10	0.45
Observations	8,915	8,913	8,667	8,915	8,913	8,667
Programs	979	979	950	979	979	950
Markets	134	221	631	134	221	631
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: The dependent variable is first-year enrollment of incumbent programs. Exposure is cumulative competitor seats added since 2010, in 10 percentage-point units of pre-period market enrollment. All regressions control for own shocks and absorb program and corresponding field × year fixed effects; the last three columns add region × year fixed effects. Standard errors clustered by market. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$. Coefficients are positive and mostly insignificant: no evidence that competitor expansions take students away.

$$\log N_{kt}^{\text{first}} = \beta_w \log \widetilde{\text{cum} E_{kt}^w} + \theta_w D_{kt}^w + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}, \quad D_{kt}^w = \mathbf{1}\{\text{cum} E_{kt}^w > 0\}$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i> <i>F</i>	0.021 (0.020) 1.07	0.026 (0.024) 1.12	-0.019 (0.030) 0.38	0.008 (0.013) 0.40	0.032* (0.016) 3.73	-0.009 (0.030) 0.10
<i>Triangular</i> <i>F</i>	0.001 (0.016) 0.00	0.010 (0.018) 0.31	-0.013 (0.026) 0.26	0.006 (0.017) 0.13	0.023 (0.015) 2.53	-0.005 (0.026) 0.04
<i>Gaussian</i> <i>F</i>	0.013 (0.014) 0.87	0.015 (0.016) 0.89	0.014 (0.020) 0.48	0.009 (0.013) 0.43	0.018 (0.016) 1.30	0.019 (0.020) 0.91
Observations	8,865	8,863	8,614	8,865	8,863	8,614
Programs	975	975	947	975	975	947

Notes: Each coefficient is β_w , the elasticity of enrollment with respect to cumulative exposure among exposed program-years. Zero-exposure observations are kept by setting $\log \widetilde{\text{cum} E} = 0$ and including D_{kt}^w , not reported. Standard errors clustered by market. Observations with zero enrollment are excluded. All elasticities are below 0.035 in absolute value.

Design III: kernel-weighted denominator

$$E_{kt}^{w,KD} = 100 \times \frac{\sum_{j \in m(k), j \neq k} w(k, j) \text{seats}_{jt}}{\sum_{\ell \in m(k)} w(k, \ell) \bar{N}_{\ell, 2007-2009}^{\text{first}}}, \quad w \in \{\text{Tri}, \text{Gau}\}, \quad w(k, k) = 1.$$

- Same numerator as before; the denominator is now the enrollment of programs with similar selectivity to k , not total market enrollment. As in the Design II kernel-denominator measure: for a selective k , the relevant market is the selective programs in $m(k)$.
- The measure keeps the scale of the shock (unlike Q) and lets its relevance depend on k 's position in the market.

Cumulative exposure in 2016, by field definition (pp)

Field definition	Exposure measure	Mean	Std. dev.	p90	p99	Max	Share zero
Broad	Total	5.79	12.27	13.10	25.54	298.01	38.2 %
	Triangular KD	4.87	7.11	12.88	31.24	67.51	43.6 %
	Gaussian KD	5.55	8.46	12.55	29.32	150.50	38.2 %
ISCED-97	Total	5.25	7.50	16.67	31.78	62.50	47.0 %
	Triangular KD	4.55	7.81	13.43	32.70	67.51	53.0 %
	Gaussian KD	5.09	7.54	13.34	30.90	60.85	47.0 %
Generic	Total	2.88	8.45	11.90	41.47	100.00	82.7 %
	Triangular KD	1.54	6.13	3.33	27.10	96.42	86.8 %
	Gaussian KD	2.48	7.79	9.01	38.49	99.94	82.7 %

Notes: Incumbent programs, cumulative exposure in 2016, zeros included. The kernel denominator does not generate extreme values: the maximum is below that of Total exposure in the Broad and ISCED-97 definitions.

Design III: results with kernel-weighted denominator

Levels: first-year enrollment, 10 pp units

Exposure measure	Field $\times$ year FE			+ Region $\times$ year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.588 (0.677)	2.413** (0.945)	1.109 (0.874)	0.368 (0.362)	1.499* (0.786)	-0.286 (0.815)
Wald F	0.76	6.51	1.61	1.03	3.64	0.12
<i>Triangular KD</i>	1.591 (1.165)	1.359* (0.724)	0.956 (0.971)	0.691 (1.150)	0.572 (0.611)	-0.741 (0.954)
Wald F	1.86	3.52	0.97	0.36	0.88	0.60
<i>Gaussian KD</i>	1.436 (1.014)	2.355*** (0.821)	1.367 (0.838)	0.829 (0.659)	1.425** (0.622)	-0.176 (0.796)
Wald F	2.01	8.23	2.66	1.58	5.25	0.05
Observations	8,915	8,913	8,667	8,915	8,913	8,667
Programs	979	979	950	979	979	950
Markets	134	221	631	134	221	631
Region $\times$ year FE	No	No	No	Yes	Yes	Yes

Logs: elasticity of first-year enrollment, zero-exposure indicator included

Exposure measure	Field $\times$ year FE			+ Region $\times$ year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.021 (0.020)	0.026 (0.024)	-0.019 (0.030)	0.008 (0.013)	0.032* (0.016)	-0.009 (0.030)
Wald F	1.07	1.12	0.38	0.40	3.73	0.10
<i>Triangular KD</i>	0.001 (0.015)	0.007 (0.017)	-0.037 (0.027)	-0.004 (0.016)	0.013 (0.015)	-0.028 (0.028)
Wald F	0.01	0.19	1.88	0.07	0.80	1.05
<i>Gaussian KD</i>	0.015 (0.013)	0.015 (0.016)	0.004 (0.023)	0.005 (0.011)	0.013 (0.015)	0.008 (0.023)
Wald F	1.36	0.87	0.03	0.22	0.75	0.13
Observations	8,865	8,863	8,614	8,865	8,863	8,614
Programs	975	975	947	975	975	947
Markets	133	219	629	133	219	629
Region $\times$ year FE	No	No	No	Yes	Yes	Yes

Notes: Program FE, corresponding field $\times$ year FE, own shocks; region $\times$ year FE where indicated. SE clustered by market. Total repeated for reference. Unlike Design II, where the kernel denominator turns the weighted measures negative and significant, here they stay positive: significant only for ISCED-97 in levels (Gaussian KD: 2.36*** and 1.43**), and all elasticities are below 0.04 in absolute value.

Design III: kernel-weighted denominator, influence of the tail

Sample	Broad		ISCED-97	
	Triangular KD	Gaussian KD	Triangular KD	Gaussian KD
All programs	0.691 (1.150)	0.829 (0.659)	0.572 (0.611)	1.425** (0.622)
Observations	8,915	8,915	8,913	8,913
Excluding top 1 % of exposure	0.191 (1.147)	1.911* (1.148)	1.043 (1.026)	2.172** (0.842)
Observations	8,847	8,845	8,845	8,833
Excluding top 5 % of exposure	-2.048 (1.824)	0.688 (1.597)	-0.466 (1.307)	1.937 (1.353)
Observations	8,530	8,529	8,516	8,525

Notes: Levels, program, field $\times$ year and region $\times$ year FE, own shocks; SE clustered by market. Programs are dropped when their cumulative KD exposure in 2016 is above the 99th or 95th percentile. Estimates move around with the trimming, and only the Gaussian ISCED-97 estimate is significant in more than one sample; nothing points to a negative effect.

Design III: decomposing weighted exposure

$$\text{cum}E_{kt}^w = \underbrace{\frac{\sum_{s \leq t} \sum_{j \in m(k)} \text{seats}_{js}}{T_{m(k)}}}_{M_{kt}: \text{seats added in market}} \times \underbrace{\frac{\sum_{s \leq t} \sum_{j \in m(k)} w(k, j) \text{seats}_{js}}{\sum_{s \leq t} \sum_{j \in m(k)} \text{seats}_{js}}}_{Q_{kt}^w: \text{similarity with shocked programs}}$$

Size of competitor expansion

M_{kt} is Total exposure: the seats added by shocks in k 's market up to t , relative to market size. It is (almost) common to all incumbents in the market.

Conditional similarity

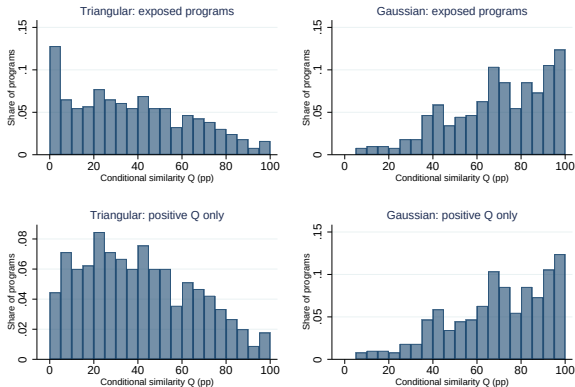
$Q_{kt}^w \in [0, 1]$ is the seat-weighted PSU similarity between k and the programs that expanded. Conditional on the size of the expansion, it captures competitive proximity.

Using similarity as the independent variable

$$N_{kt}^{\text{first}} = \beta_w Q_{kt}^w + \lambda \mathbf{1}\{M_{kt} > 0\} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}, \quad M_{k,2016} > 0.$$

The sample is restricted to Broad-field markets with at least one shock. $Q_{kt}^w = 0$ before the first shock in the market, and $\mathbf{1}\{M_{kt} > 0\}$ absorbs the level effect of that first shock. β_w is the enrollment change associated with a 0.10 increase in similarity with the programs that expanded.

Design III: distribution of conditional similarity



Incumbent programs in broad-field markets with at least one shock by 2016; Q in 2016.

- Incumbent programs in Broad-field markets with at least one shock by 2016; Q in 2016, in percentage points.
- As in Design II, the triangular measure piles up near zero and the Gaussian measure is more dispersed.

$$N_{kt}^{\text{first}} = \beta_w Q_{kt}^w + \lambda \mathbf{1}\{M_{kt} > 0\} + \dots, \quad \log N_{kt}^{\text{first}} = \delta_w \log \widetilde{Q_{kt}^w} + \theta_w \mathbf{1}\{Q_{kt}^w > 0\} + \lambda \mathbf{1}\{M_{kt} > 0\} + \dots$$

	Levels		Logs	
	Triangular	Gaussian	Triangular	Gaussian
Similarity term	-0.394 (0.240)	-0.316 (0.239)	-0.014 (0.022)	-0.003 (0.027)
Wald F	2.70	1.74	0.40	0.01
Observations	5,576	5,576	5,562	5,562
Programs	592	592	592	592
Markets	46	46	46	46

Notes: Broad-field markets with at least one shock by 2016. Levels: coefficient per 0.10 increase in Q . Logs: elasticity with respect to Q , with zero- Q observations kept through the indicator. All regressions control for own shocks and absorb program, Broad-field $\times$ year and region $\times$ year fixed effects. Standard errors clustered by market. The sign matches Design II (closer competitors, lower enrollment), but no estimate is significant.

Design III: heterogeneity by incumbent selectivity

	PSU < 550		PSU 550–649		PSU ≥ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
<i>Vacancy-shock exposure</i>						
Coefficient	3.60 (2.73)	2.74 (1.92)	0.06 (1.23)	-0.01 (0.42)	-23.59 (28.11)	-13.51 (17.80)
F-statistic	1.74	2.03	0.00	0.00	0.70	0.58
Observations	2,785		4,783		1,314	
Programs	335		502		139	
Clusters	95		98		25	
<i>Conditional similarity Q</i>						
Coefficient	-0.13 (0.36)	-0.35* (0.20)	-0.21 (0.31)	-0.05 (0.46)	-2.18 (2.26)	-4.04 (3.81)
F-statistic	0.13	2.96	0.44	0.01	0.94	1.13
Observations	1,315		3,239		779	
Programs	148		338		83	
Clusters	24		34		10	

Notes: Incumbent PSU is the 2007–2009 mean entrant PSU. Standard errors clustered by Broad-field market. The first panel includes program, Broad-field × year and region × year FE; the second includes program and market × year FE and is restricted to markets with at least one shock. Coefficients correspond to a 10 percentage-point increase in exposure or a 0.10 increase in conditional similarity. Only Gaussian Q among low-PSU programs is significant (10 %); estimates for PSU ≥ 650 are very imprecise.

Design III: cosine exposure

Each program is represented by the distribution of its first-year students across 25-point PSU bins (2007–2009 entrants; first three cohorts for later programs), as in Design II.

1. Similarity between incumbent k and source j

$$\cos \theta_{j,k}^{PSU} = \frac{\mathbf{v}_j \cdot \mathbf{v}_k}{\|\mathbf{v}_j\| \|\mathbf{v}_k\|}, \quad \cos \theta^{PSU \times R} = \cos \theta^{PSU} \mathbf{1}\{r_j = r_k\}, \quad \cos \theta^{PSU \times R \times F} = \cos \theta^{PSU} \mathbf{1}\{r_j = r_k, f_j = f_k\},$$

$$\cos \theta^{PSU+R} = \frac{\cos \theta^{PSU} + \mathbf{1}\{r_j=r_k\}}{2}, \quad \cos \theta^{PSU+R+F} = \frac{\cos \theta^{PSU} + \mathbf{1}\{r_j=r_k\} + \mathbf{1}\{f_j=f_k\}}{3}.$$

Each program has one campus region and one Broad field, so the interaction-cell cosines equal the PSU cosine when programs share the region (and field), and zero otherwise.

2. Exposure of incumbent program k

$$E_{kt}^d = \sum_{j \neq k} w_{jt} \cos \theta_{j,k}^d, \quad w_{jt} = \frac{\text{seats}_{jt}}{\sum_{s=2010}^{2016} \sum_{\ell} \text{seats}_{\ell s}}, \quad \text{cum} E_{kt}^d = \sum_{s=2010}^t E_{ks}^d.$$

The weights sum to one over all 3,307 seats added by shocks in 2010–2016, as entrant enrollment weights sum to one in Design II. In 2016, $\text{cum} E_k^d$ is k 's seat-weighted similarity with every program that expanded.

Design III: cosine exposure, samples and descriptive statistics

Panel A: Analytical samples

Variable	Full sample	$\bar{N}_{k,2007-09}^{\text{first}} \geq 10$
Programs	989	943
Program-year observations	8,925	8,647
Incumbent universities	25	25
Mean annual enrollment, 2007–2009	51.7	53.9
Mean PSU score, 2007–2009	583.9	586.4

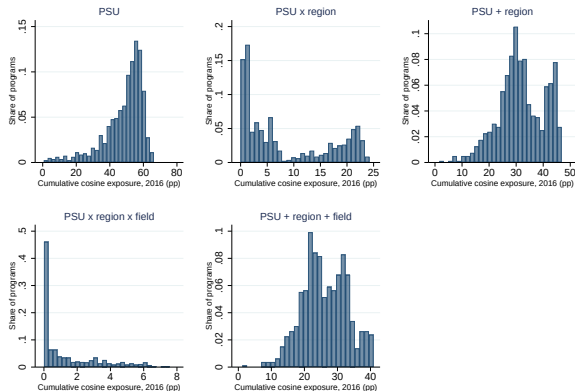
Panel B: Cumulative cosine exposure in 2016

Exposure measure	Full sample			$\bar{N}_k^{\text{first}} \geq 10$		
	Mean	Std. dev.	Share zero	Mean	Std. dev.	Share zero
PSU bins	0.4834	0.1203	0.0 %	0.4862	0.1183	0.0 %
PSU bins $\times$ campus region	0.0858	0.0828	6.3 %	0.0876	0.0829	6.0 %
PSU bins + campus region	0.3182	0.0851	0.0 %	0.3211	0.0831	0.0 %
PSU bins $\times$ campus region $\times$ Broad field	0.0135	0.0180	38.2 %	0.0137	0.0181	37.9 %
PSU bins + campus region + Broad field	0.2619	0.0683	0.0 %	0.2642	0.0670	0.0 %

Notes: The unit is an incumbent program; moments are unweighted across programs. The restricted sample excludes programs with fewer than 10 first-year students per year in 2007–2009; the restriction is applied after exposure construction. Multiplicative measures require coincidence in the indicated dimensions; additive measures give equal weight to PSU similarity, campus-region coincidence and, where applicable, Broad-field coincidence.

Design III: distribution of cosine exposure

Positive exposure only →



Incumbent programs, full sample. All programs.

- Each incumbent program appears once, at its cumulative cosine exposure in 2016, in percentage points.
- As in Design II, the multiplicative $\text{PSU} \times \text{region} \times \text{field}$ measure has the most mass at zero and the smallest support.

$$N_{kt}^{\text{first}} = \beta_d \text{cum} E_{kt}^d + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}$$

Exposure measure	Full sample			$\bar{N}_k^{\text{first}} \geq 10$		
	(1) + year	(2) + field × year	(3) + region × year	(4) + year	(5) + field × year	(6) + region × year
PSU bins	-3.852* (2.082)	-3.963* (2.330)	-2.447 (2.355)	-4.003* (2.197)	-4.076 (2.482)	-2.230 (2.476)
Wald F	3.42	2.89	1.08	3.32	2.70	0.81
PSU bins × region	2.136*** (0.376)	2.305*** (0.392)	-3.285 (2.608)	2.133*** (0.390)	2.344*** (0.409)	-3.267 (2.645)
Wald F	32.21	34.55	1.59	29.91	32.86	1.53
PSU bins × region × field	1.584*** (0.383)	1.969*** (0.476)	-0.334 (0.890)	1.557*** (0.392)	1.965*** (0.493)	-0.393 (0.912)
Wald F	17.09	17.12	0.14	15.81	15.86	0.19
PSU bins + region	2.629** (1.139)	2.938** (1.289)	-3.296 (3.048)	2.701** (1.236)	3.164** (1.417)	-3.015 (3.218)
Wald F	5.33	5.20	1.17	4.77	4.99	0.88
PSU bins + region + field	2.829*** (0.864)	3.434** (1.524)	-4.000 (3.553)	2.903*** (0.923)	3.767** (1.703)	-3.673 (3.812)
Wald F	10.72	5.08	1.27	9.89	4.89	0.93
Observations	8,915	8,915	8,915	8,640	8,640	8,640
Programs	979	979	979	936	936	936

Notes: Exposure in SD units; program-clustered standard errors. Columns (1) and (4): program and year FE; (2) and (5): Broad-field × year FE; (3) and (6): add campus-region × year FE. Columns (4)–(6) exclude programs with fewer than 10 first-year students per year in 2007–2009. The pattern replicates Design II: measures that include region are positive and strongly significant until region × year FE are added, and then turn negative and insignificant. The PSU-only measure is negative (−3.9, 10 %) without region × year FE.

Design III: robustness to the shock definition

	Main	Top 10 %	Top 25 %	Within-year	No sudden	Levels	INDICES	Admissions
<i>Total</i>	0.368 (0.362)	0.269 (0.289)	0.230 (0.317)	0.440 (0.414)	0.471 (0.415)	0.352 (0.387)	0.180 (0.270)	0.964 (0.989)
<i>Gaussian</i>	0.673 (0.654)	0.657 (0.710)	0.614 (0.654)	0.866 (0.725)	0.765 (0.648)	1.069 (0.814)	-0.054 (0.486)	-0.774 (1.348)
<i>Gaussian KD</i>	0.829 (0.659)	1.012 (0.820)	0.832 (0.673)	1.119 (0.809)	0.855 (0.634)	1.517 (0.995)	0.189 (0.447)	-0.782 (1.391)
Observations	8,915	8,915	8,915	8,915	8,915	8,915	8,915	8,915

Definition		Shocks	Mean $\Delta \log V$	Mean seats added
Main	top 20 %, sudden	179	0.39	18.5
Top 10 %	sudden	79	0.50	24.4
Top 25 %	sudden	205	0.37	18.8
Within-year p80	sudden	188	0.38	18.3
No sudden conditions	top 20 %	220	0.39	18.5
Level change	top 20 % of ΔV	152	0.35	25.4
INDICES vacancies	top 20 %, sudden	127	0.40	21.1
Admissions as vacancies	top 20 %, sudden	319	0.52	16.8

Notes: Specification (1), Broad-field markets, with program, field $\times$ year and region $\times$ year FE and own shocks; standard errors clustered by market. Exposure rebuilt under each shock definition; Gaussian KD uses the kernel-weighted denominator. No definition delivers a significant effect.

Design III: event studies, cohorts and pre-trend tests

Treatment cohorts (programs in 2016)

First treatment year	Market shock	Close shock	Top-5 cosine shock
Never treated	304	347	460
2010	235	153	51
2011	130	136	70
2012	50	42	28
2013	21	48	34
2014	16	23	44
2015	11	13	45
2016	29	34	64
Programs	796	796	796

Joint pre-trend tests, p -values

Treatment	Outcome	TWFE	Callaway-Sant'Anna	
			Never treated	Not yet treated
Market shock	Enrollment	0.348	0.080	0.081
	Log enrollment	0.124	0.011	0.012
Close-competitor shock	Enrollment	0.539	0.000	0.000
	Log enrollment	0.240	0.000	0.000
Top-5 cosine shock	Enrollment	0.293	0.021	0.021
	Log enrollment	0.150	0.068	0.069

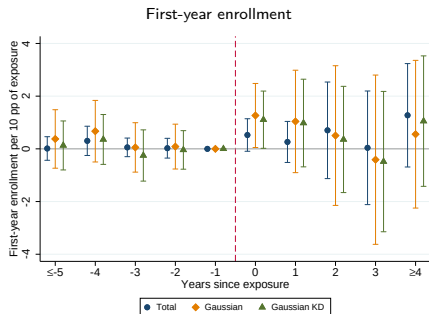
- Callaway & Sant'Anna is estimated twice: with never-treated programs as the only controls, and with not-yet-treated plus never-treated programs. Between 38 % and 58 % of programs are never treated, so both are feasible; they give nearly identical estimates.
- TWFE (with field- and region-year FE) never rejects parallel pre-trends. CS rejects for the close-competitor shock and, at 5 %, for log enrollment after a market shock and enrollment after a cosine shock.
- After a close-competitor or market shock, CS shows enrollment *rising* by 2–4 students; TWFE, which absorbs field- and region-year shocks, is flat. After a shock at a top-5 cosine program, both point to declines of 1–3 students, imprecisely estimated.
- These event studies use a binary first-shock indicator and discard the dose. The next two slides keep it.

Design III: event studies with the treatment dose

Distributed-lag event study with binned endpoints (Schmidheiny & Siegloch 2023) on the yearly exposure flow E_{kt} (10 pp):

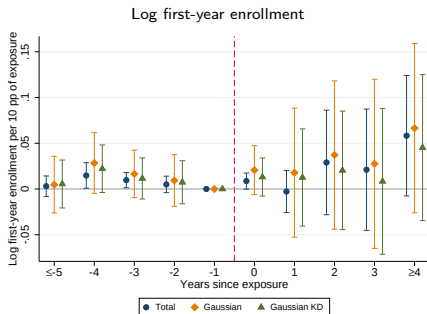
$$y_{kt} = \beta_{\leq -5} \sum_{s \geq t+5} E_{ks} + \sum_{l=-4}^{-2} \beta_l E_{k,t-l} + \sum_{l=0}^3 \beta_l E_{k,t-l} + \beta_{\geq 4} \sum_{s \leq t-4} E_{ks} + \delta \text{Own}_{kt} + \mu_k + \alpha_{f(k),t} + \gamma_{r(k),t} + \varepsilon_{kt}$$

$E_{k,t+1}$ is omitted, so β_l is the cumulative effect at event time l of 10 pp of exposure received at $l = 0$, relative to $l = -1$.



Pretrend p-values: Total 0.604; Gaussian 0.668; Gaussian KD 0.604.

Distributed-lag event study, binned endpoints, $t = -1$ omitted; program, field-year and region-year FE, own shocks.



Pretrend p-values: Total 0.102; Gaussian 0.284; Gaussian KD 0.325.

Distributed-lag event study, binned endpoints, $t = -1$ omitted; program, field-year and region-year FE, own shocks.

Notes: Broad field $\times$ region market; program, field $\times$ year and region $\times$ year FE, own shocks; SE clustered by market. Leads beyond 2016 set to zero, as in the binary event studies.

Design III: dose event studies, average post-exposure effect

	Enrollment		Log enrollment	
	(1)	(2)	(3)	(4)
<i>A. All years (2007–2016)</i>				
Total	0.890 (0.665) [0.524]	0.383 (0.547) [0.604]	0.0286 (0.0228) [0.083]	0.0140 (0.0162) [0.102]
Gaussian	0.928 (0.979) [0.824]	0.600 (0.940) [0.668]	0.0318 (0.0337) [0.618]	0.0258 (0.0274) [0.284]
Gaussian KD	0.978 (0.763) [0.733]	0.490 (0.762) [0.604]	0.0258 (0.0268) [0.610]	0.0136 (0.0218) [0.325]
Triangular KD	0.786 (1.142) [0.352]	0.376 (1.183) [0.608]	0.0078 (0.0315) [0.145]	0.0002 (0.0290) [0.881]
Observations	8,915	8,915	8,865	8,865
<i>B. Years 2007–2012 (all leads observed)</i>				
Total	1.170 (1.506) [0.382]	1.795 (1.404) [0.091]	0.0537 (0.0439) [0.310]	0.0251 (0.0403) [0.055]
Gaussian	0.767 (1.837) [0.381]	2.308 (1.995) [0.200]	0.0312 (0.0608) [0.203]	0.0255 (0.0622) [0.162]
Gaussian KD	0.838 (1.457) [0.371]	1.773 (1.654) [0.450]	0.0263 (0.0445) [0.174]	0.0072 (0.0466) [0.291]
Triangular KD	0.046 (1.306) [0.821]	0.653 (1.424) [0.018]	-0.0070 (0.0422) [0.198]	-0.0173 (0.0453) [0.933]
Observations	5,555	5,555	5,532	5,532
Field-year FE	Yes	Yes	Yes	Yes
Region-year FE	No	Yes	No	Yes

Notes: Coefficient: average of $\beta_0, \dots, \beta_3$ per 10 pp of exposure (SE in parentheses); p -value of the joint test

$\beta_{-5} = \beta_{-4} = \beta_{-3} = \beta_{-2} = 0$ in brackets. Panel B keeps years up to 2012, where every interior lead is observed. Program FE, field $\times$ year FE, own shocks; SE clustered by market.

- Keeping the dose, pre-trends are not rejected at 5 % for any measure in the full sample, unlike the binary CS event studies (which also lack field- and region-year FE). In panel B, one of 16 tests rejects (Triangular KD, $p = 0.018$).
- Post-exposure effects are positive and insignificant (0.4–1 students per 10 pp; below 3 % in logs), consistent with the static first stages. No sign of business stealing.

Appendix

Non-harmonized field observations

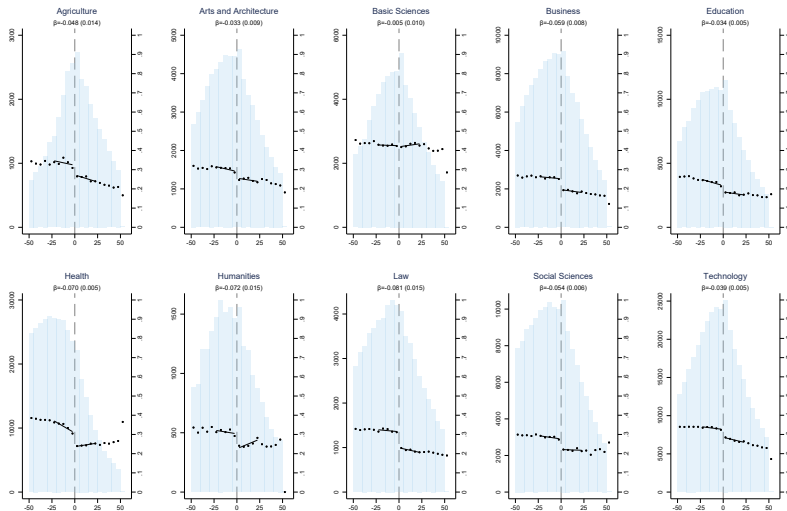
Institution	Program	Years	Code	Obs.
UV	Bachillerato ingreso común	2007–2008	1900	558
USACH	Tecnólogo en alimentos (vesp)	2007–2010	1690	208
UV	Administración pública – diurno	2007	1997	26
UV	Actuación teatral	2008	1928	9
Total		2007–2010		801

These observations are excluded from the field-enriched sample because their program field could not be reliably harmonized. They represent 0.095 % of the baseline RDD sample within the main bandwidth.

[← Back](#)

Appendix note 2: PSU retakes by field

RDD: Retakes PSU Within Two Years by Field
Outcome: retakes_psu_2y. Regression BW = ± 25 ; plot range = ± 50

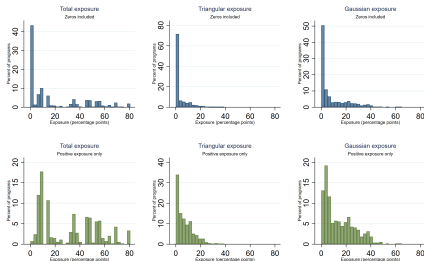


The reduction in PSU retaking is generally negative across fields, although magnitudes vary by academic area. [← Back](#)

SUA exposure: alternative field definitions

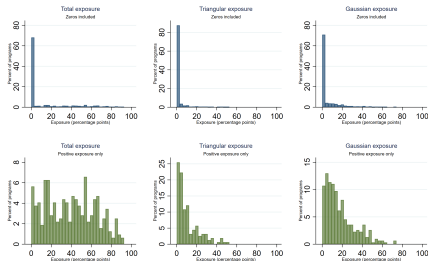
← Broad field

SUA exposure distributions: ISCED-97 field
One observation per incumbent program



Top row includes zeros; bottom row conditions on positive exposure.

SUA exposure distributions: Generic field
One observation per incumbent program



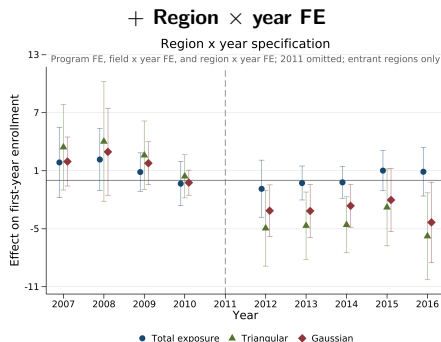
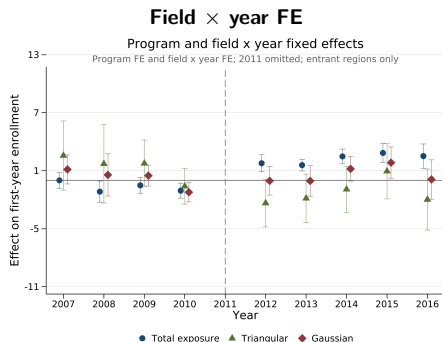
Top row includes zeros; bottom row conditions on positive exposure.

Narrower field definitions generate more markets and a larger share of zero-exposure programs, particularly under the Generic-field definition.

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times \text{Post}_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	2.807*** (0.457)	2.609*** (0.415)	2.378*** (0.381)	-0.492 (0.940)	0.652 (0.573)	0.151 (0.504)
Wald <i>F</i>	37.81	39.49	38.90	0.27	1.30	0.09
Programs with zero exposure	10	60	256	10	60	256
<i>Triangular</i>	-2.133 (1.482)	-0.830 (1.253)	0.711 (1.071)	-6.390** (2.434)	-4.179*** (1.581)	-1.575 (1.033)
Wald <i>F</i>	2.07	0.44	0.44	6.89	6.99	2.33
Programs with zero exposure	96	192	416	96	192	416
<i>Gaussian</i>	0.545 (0.892)	1.118 (0.764)	1.652** (0.725)	-4.168** (1.863)	-2.416** (1.177)	-1.240 (0.773)
Wald <i>F</i>	0.37	2.14	5.19	5.01	4.21	2.58
Programs with zero exposure	10	60	267	10	60	267
Observations	5,090	5,080	4,781	5,090	5,080	4,781
Programs	601	601	571	601	601	571
Markets	40	80	303	40	80	303
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Same specification as the main first stage. Exposure in 10 percentage-point units; SE clustered by pre-treatment market. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.



Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p -values (2007–2010) for Total, Triangular and Gaussian are 0.046, 0.068 and 0.010 in the left panel, and 0.151, 0.302 and 0.209 in the right panel.

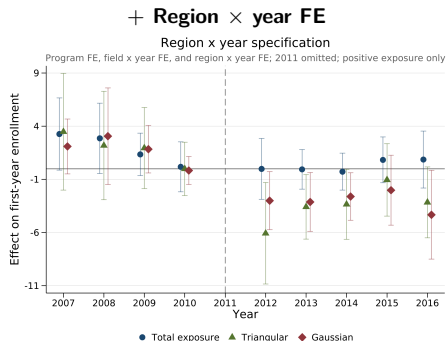
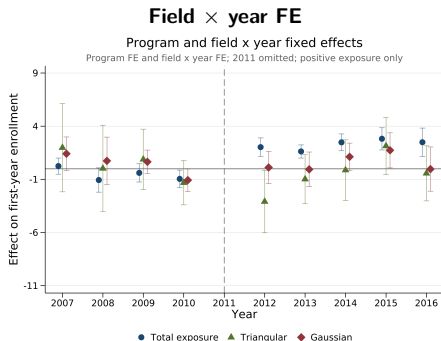
$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times \text{Post}_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	2.753*** (0.474)	2.423*** (0.443)	2.579*** (0.575)	-0.863 (0.952)	1.086 (0.911)	-1.266 (1.144)
Wald F	33.68	29.84	20.11	0.82	1.42	1.23
Programs	591	540	304	591	540	304
<i>Triangular</i>	-0.668 (1.454)	0.555 (1.264)	0.385 (1.496)	-4.698** (2.119)	-2.671 (1.678)	-1.705 (1.624)
Wald F	0.21	0.19	0.07	4.91	2.54	1.10
Programs	505	407	142	505	407	142
<i>Gaussian</i>	0.378 (0.908)	0.805 (0.798)	0.489 (0.874)	-4.196** (1.865)	-2.576* (1.289)	-2.133** (0.897)
Wald F	0.17	1.02	0.31	5.06	4.00	5.65
Programs	591	540	292	591	540	292
Observations (Total)	5,009	4,571	2,516	5,009	4,571	2,516
Programs (Total)	591	540	304	591	540	304
Markets (Total)	36	59	131	36	59	131
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Same specification as the main first stage. Exposure in 10 percentage-point units; SE clustered by pre-treatment market. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Event studies: positive exposure only

← First stage



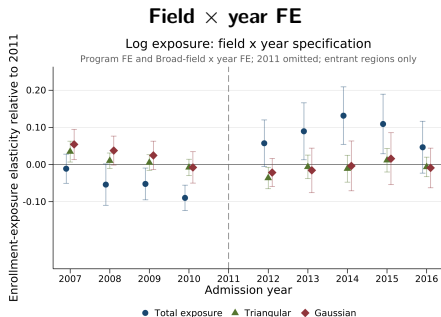
Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p -values (2007–2010) for Total, Triangular and Gaussian are 0.072, 0.055 and 0.012 in the left panel, and 0.132, 0.348 and 0.207 in the right panel.

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^k} \times \text{Post}_t \right] + \theta_k \left[D_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

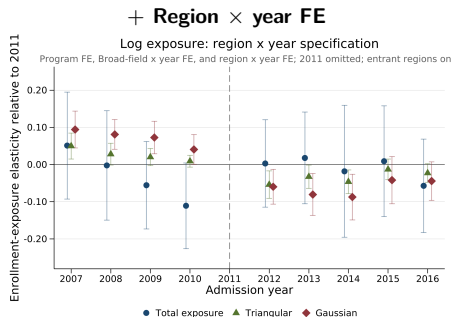
Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.133*** (0.039)	0.109*** (0.035)	0.059* (0.031)	0.025 (0.086)	0.012 (0.041)	0.006 (0.031)
Wald F	11.66	9.63	3.51	0.09	0.09	0.04
Programs with zero exposure	10	60	256	10	60	256
<i>Triangular</i>	-0.015 (0.017)	0.006 (0.018)	0.029 (0.022)	-0.052*** (0.018)	-0.023 (0.017)	0.005 (0.018)
Wald F	0.77	0.10	1.70	8.22	1.84	0.09
Programs with zero exposure	96	192	416	96	192	416
<i>Gaussian</i>	-0.024 (0.036)	-0.009 (0.034)	0.019 (0.019)	-0.117*** (0.034)	-0.075** (0.031)	-0.021 (0.017)
Wald F	0.44	0.06	0.96	11.57	5.83	1.46
Programs with zero exposure	10	60	267	10	60	267
Observations	5,090	5,080	4,781	5,090	5,080	4,781
Programs	601	601	571	601	601	571
Markets	40	80	303	40	80	303
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Same specification as the main log first stage: $\widetilde{\log E_p^k} = \log E_p^k$ if $E_p^k > 0$ and 0 otherwise, $D_p^k = \mathbf{1}\{E_p^k > 0\}$. Observations with zero first-year enrollment are excluded. SE clustered by pre-treatment market. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

Log event studies: regions with entrant universities only ← First stage



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
joint pretrend p-values: Total = 0.000; Triangular = 0.017; Gaussian = 0.021.



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
joint pretrend p-values: Total = 0.015; Triangular = 0.051; Gaussian = 0.000.

Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Plotted coefficients are those on the log-exposure × year terms, relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular and Gaussian are 0.000, 0.018 and 0.021 in the left panel, and 0.015, 0.051 and 0.000 in the right panel.

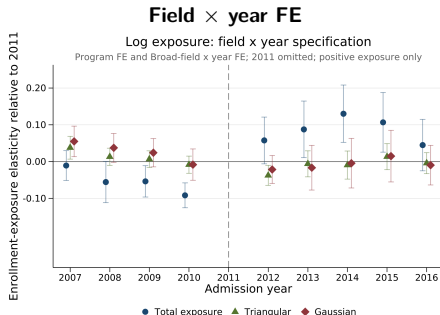
$$\log N_{pt}^{\text{first}} = \beta_k [\log E_p^k \times \text{Post}_t] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \quad E_p^k > 0.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.132*** (0.039)	0.126*** (0.036)	0.134*** (0.021)	0.019 (0.086)	0.046 (0.061)	0.100*** (0.024)
Wald F	11.39	12.01	41.28	0.05	0.58	17.13
Programs	591	540	304	591	540	304
<i>Triangular</i>	-0.015 (0.019)	0.004 (0.022)	0.021 (0.027)	-0.056*** (0.020)	-0.028 (0.021)	0.004 (0.023)
Wald F	0.61	0.03	0.56	8.06	1.84	0.03
Programs	505	407	142	505	407	142
<i>Gaussian</i>	-0.025 (0.037)	-0.004 (0.036)	0.018 (0.022)	-0.117*** (0.034)	-0.078** (0.035)	-0.026 (0.020)
Wald F	0.46	0.01	0.64	11.55	4.97	1.79
Programs	591	540	292	591	540	292
Observations (Total)	5,009	4,571	2,516	5,009	4,571	2,516
Programs (Total)	591	540	304	591	540	304
Markets (Total)	36	59	131	36	59	131
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

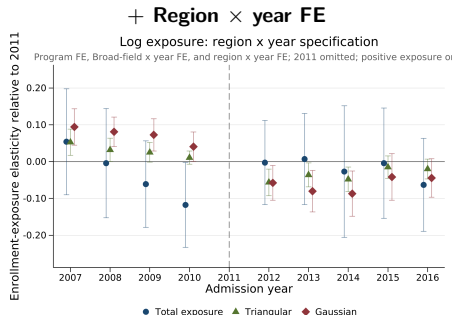
Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Log first stage without zero-exposure programs, so $D_p^k \times \text{Post}_t$ equals Post_t and is absorbed by the fixed effects. Observations with zero first-year enrollment are excluded. SE clustered by pre-treatment market. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Log event studies: positive exposure only

← First stage



Zero-exposure programs excluded; D x year controls absorbed by the fixed effects.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.000; Triangular = 0.029; Gaussian = 0.022.

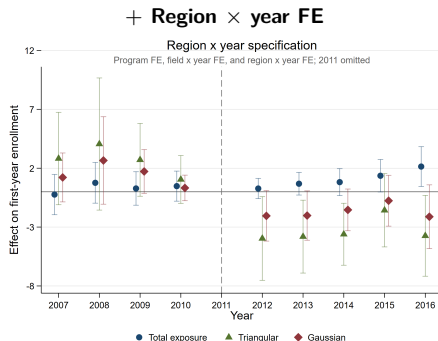
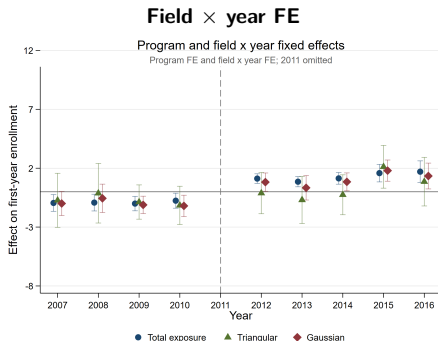


Zero-exposure programs excluded; D x year controls absorbed by the fixed effects.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.009; Triangular = 0.039; Gaussian = 0.000.

Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Plotted coefficients are those on the log-exposure × year terms, relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular and Gaussian are 0.000, 0.029 and 0.022 in the left panel, and 0.009, 0.039 and 0.000 in the right panel.

SUA exposure: event-study estimates

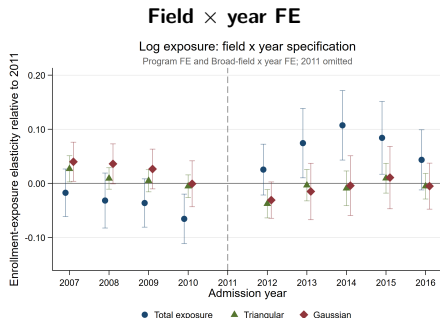
← First stage



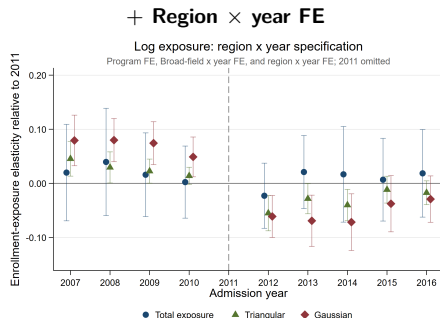
Notes: The figures use the Broad-field market definition and report separate event studies for Total, Triangular and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects. The right panel additionally absorbs pre-treatment region × year fixed effects. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011, the omitted year. Vertical bars report 95 % confidence intervals and standard errors are clustered by the pre-treatment market. Joint pretrend p -values for Total, Triangular and Gaussian exposure are 0.024, 0.568 and 0.020 in the left panel, and 0.285, 0.385 and 0.368 in the right panel.

SUA exposure: log event-study estimates

← Log first stage



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.034; Triangular = 0.101; Gaussian = 0.100.



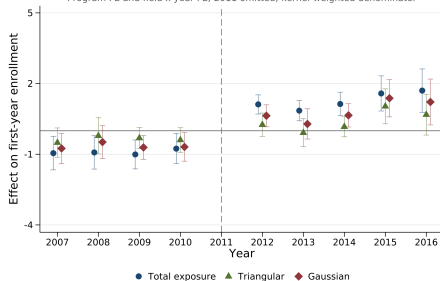
Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.857; Triangular = 0.067; Gaussian = 0.000.

Notes: The figures use the Broad-field definition and report separate event studies for Total, Triangular, and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects; the right panel additionally absorbs pre-treatment region × year fixed effects. For each exposure measure, zero-exposure programs are retained by setting $\log \widetilde{E_p^k} = 0$ when $E_p^k = 0$ and including $\mathbf{1}\{E_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately for every event year. The plotted coefficients correspond to $\log \widetilde{E_p^k} \times \mathbf{1}\{t = s\}$ and are interpreted relative to 2011, the omitted year. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. Vertical bars report 95 % confidence intervals; standard errors are clustered by the pre-treatment market. Joint tests of the 2007–2010 log-exposure coefficients yield pretrend p -values of 0.034, 0.101, and 0.100 for Total, Triangular, and Gaussian exposure in the left panel, and 0.857, 0.067, and 0.0001, respectively, in the right panel.

Field × year FE

Program and field x year fixed effects

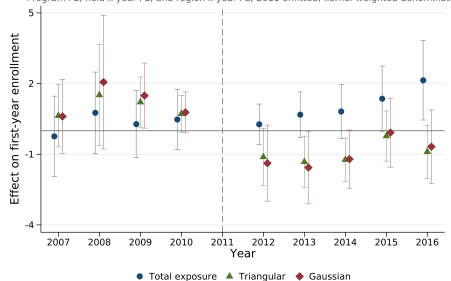
Program FE and field x year FE; 2011 omitted; kernel-weighted denominator



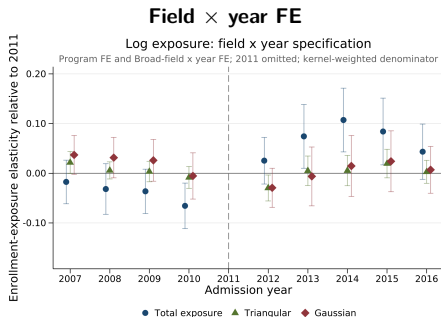
+ Region × year FE

Region x year specification

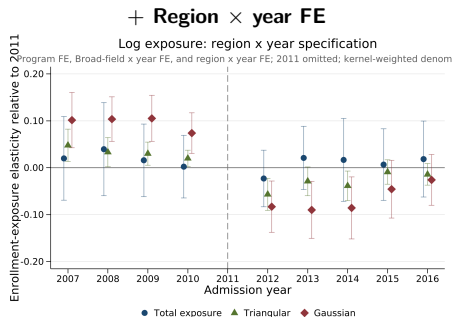
Program FE, field x year FE, and region x year FE; 2011 omitted; kernel-weighted denominator



Notes: Triangular and Gaussian exposure use the kernel-weighted denominator; Total exposure is unchanged and shown for reference. Broad-field markets; separate event studies for each measure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p -values (2007–2010) for Total, Triangular KD and Gaussian KD are 0.024, 0.170 and 0.022 in the left panel, and 0.285, 0.146 and 0.177 in the right panel.



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.034; Triangular = 0.171; Gaussian = 0.133.



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.857; Triangular = 0.056; Gaussian = 0.000.

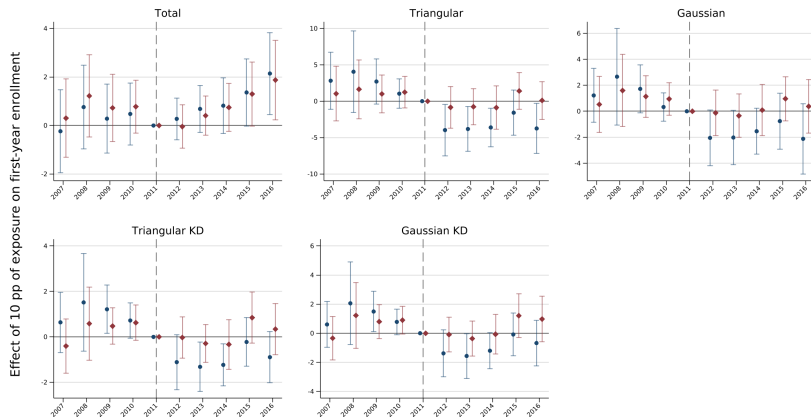
Notes: Triangular and Gaussian exposure use the kernel-weighted denominator; Total exposure is unchanged and shown for reference. Broad-field markets; separate event studies for each measure. Plotted coefficients are those on the log-exposure $\times$ year terms, relative to 2011; $1\{E > 0\} \times$ year controls included. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p -values (2007–2010) for Total, Triangular KD and Gaussian KD are 0.034, 0.171 and 0.133 in the left panel, and 0.857, 0.056 and 0.0001 in the right panel.

Exposure measure	Levels			Logs		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>A. Field × year + region × year FE</i>						
Total	0.798 (0.662)	0.904* (0.510)	0.324 (0.338)	-0.005 (0.039)	-0.024 (0.032)	-0.022 (0.027)
Triangular	-5.252** (2.115)	-3.415*** (1.310)	-0.975 (0.828)	-0.050*** (0.017)	-0.024 (0.016)	-0.005 (0.017)
Gaussian	-2.767* (1.401)	-1.447* (0.832)	-0.558 (0.571)	-0.107*** (0.030)	-0.079*** (0.026)	-0.025 (0.017)
Triangular KD	-1.726** (0.737)	-1.286*** (0.427)	-0.208 (0.470)	-0.052*** (0.018)	-0.032* (0.017)	-0.014 (0.018)
Gaussian KD	-1.919* (1.051)	-1.049* (0.600)	-0.218 (0.472)	-0.138*** (0.037)	-0.095*** (0.028)	-0.030* (0.018)
<i>B. + selectivity decile × year FE</i>						
Total	0.285 (0.569)	0.407 (0.391)	0.262 (0.333)	-0.038 (0.026)	-0.034 (0.024)	-0.019 (0.021)
Triangular	-1.125 (1.286)	-0.403 (0.962)	0.575 (0.877)	-0.016 (0.016)	-0.020 (0.015)	-0.007 (0.015)
Gaussian	-0.614 (0.949)	-0.092 (0.619)	0.308 (0.615)	-0.044** (0.017)	-0.034** (0.015)	-0.010 (0.015)
Triangular KD	-0.183 (0.451)	-0.064 (0.333)	0.588 (0.503)	-0.014 (0.017)	-0.020 (0.016)	-0.015 (0.016)
Gaussian KD	-0.210 (0.760)	0.060 (0.457)	0.327 (0.501)	-0.044** (0.020)	-0.035** (0.016)	-0.013 (0.015)
Observations	8,038	8,027	7,753	8,038	8,027	7,753

Notes: Panel A reproduces the region × year columns of the main (levels, logs) and kernel-denominator first stages. Panel B adds fixed effects for deciles of the incumbent's 2009–2011 mean PSU score interacted with year. Weighted exposure is entrant share times conditional similarity, which falls with incumbent selectivity (appendix). Levels in 10 percentage-point units; logs as in the main log first stage. SE clustered by pre-treatment market. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

SUA exposure and selectivity trends: event studies

← Table



Notes: Broad-field levels event studies of first-year enrollment on $10 \times E_p \times \mathbf{1}\{t = s\}$, 2011 omitted. Blue circles: program, field $\times$ year and region $\times$ year fixed effects. Red diamonds: adding selectivity-decile $\times$ year fixed effects. 95 % confidence intervals; SE clustered by pre-treatment market. Joint 2007–2010 pretrend p -values (blue / red): Total 0.285 / 0.385; Triangular 0.385 / 0.836; Gaussian 0.368 / 0.592; Triangular KD 0.146 / 0.219; Gaussian KD 0.177 / 0.055. Mean 2012–16 coefficient (blue / red): Triangular -3.34 / -0.19 ; Gaussian -1.70 / 0.19 .

Decomposing weighted exposure

$$E_p^k = \underbrace{\frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|) N_q}{\sum_{\ell \in m(p)} N_\ell}}_{\text{Direct construction of weighted exposure}} \equiv \frac{\sum_{q \in \mathcal{E}_{m(p)}} N_q}{\sum_{\ell \in m(p)} N_\ell} \left[\frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|) N_q}{\sum_{q \in \mathcal{E}_{m(p)}} N_q} \right]$$

$$E_p^k = \underbrace{M_{m(p)}}_{\text{entrant market share}} \times \underbrace{Q_p^k}_{\text{similarity with entrants}}$$

Entrant presence

$$M_m = \frac{\sum_{q \in \mathcal{E}_m} N_q}{\sum_{\ell \in m} N_\ell}.$$

M_m is the entrant share of total market enrollment. It captures the **size of entrant presence** and is common to all incumbent programs in market m .

Conditional similarity

$$Q_p^k = \frac{\sum_{q \in \mathcal{E}_m} k(|S_p - S_q|) N_q}{\sum_{q \in \mathcal{E}_m} N_q}.$$

Q_p^k is the entrant-enrollment-weighted PSU similarity between incumbent p and the entrant programs in its market. It is higher when larger entrants have PSU profiles closer to that of p . Conditional on entrant presence, it captures **competitive proximity** rather than the overall size of entry.

Using similarity as the independent variable

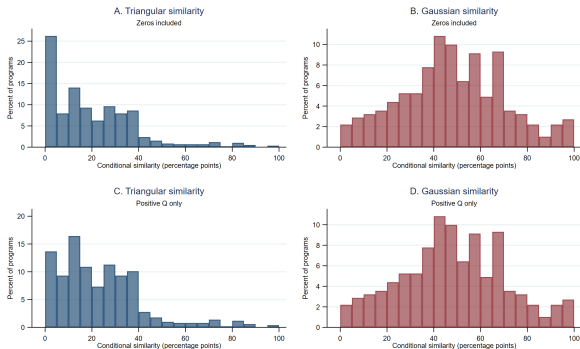
$$N_{pt}^{\text{first}} = \beta_k \left[Q_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \quad M_{m(p)} > 0.$$

The sample is restricted to markets with positive entrant presence. Program fixed effects absorb time-invariant differences across incumbents, while Broad-field $\times$ year and pre-treatment region $\times$ year fixed effects absorb common field-specific and regional shocks over time. The model therefore asks whether, among incumbents in exposed markets, programs whose pre-treatment PSU profiles are more similar to the entrants experience different post-2012 enrollment changes. Unlike E_p^k , which combines entrant market share and similarity, Q_p^k isolates the **similarity component** of exposure. The coefficient β_k measures the differential post-2012 enrollment change associated with a 0.10 increase in entrant-enrollment-weighted PSU similarity.

Distribution of conditional similarity measures

Distribution of conditional similarity measures

One observation per incumbent program in exposed markets



The upper row includes zeros; the lower row conditions on positive Q .
The horizontal axis reports conditional similarity in percentage points and the vertical axis reports the share of programs.

- Each incumbent program appears once. The horizontal axis reports conditional similarity Q in percentage points and the vertical axis the share of programs.
- The upper row includes all exposed programs, while the lower row excludes zeros and shows the distribution conditional on positive Q .
- Triangular similarity is concentrated at relatively low values and contains a relevant mass near zero. Gaussian similarity is more dispersed and shifted toward higher values, indicating a less restrictive measure of similarity across exposed markets.

Common fixed-effects specification

$$N_{pt}^{\text{first}} = \beta_k [Q_p^k \times \text{Post}_t] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \quad \log N_{pt}^{\text{first}} = \delta_k [\widetilde{\log Q_p^k} \times \text{Post}_t] + \theta_k [D_p^k \times \text{Post}_t] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log Q_p^k} = \begin{cases} \log(Q_p^k), & Q_p^k > 0, \\ 0, & Q_p^k = 0, \end{cases} \quad D_p^k = \mathbf{1}\{Q_p^k > 0\}, \quad M_m > 0.$$

	Levels		Logs	
	Triangular	Gaussian	Triangular	Gaussian
Similarity term $\times$ Post	-2,381*** (0,491)	-2,241*** (0,487)	-0,067*** (0,020)	-0,142*** (0,035)
Wald F	23.50	21.18	11.37	16.29
Observations	5,009	5,009	5,009	5,009
Programs	591	591	591	591
Markets	36	36	36	36

Notes: In levels, a 0.10 increase in conditional similarity is associated with approximately 2.2–2.4 fewer first-year students after 2012. In logs, a 10 % increase in similarity is associated with approximately 0.67 % lower enrollment under the triangular measure and 1.42 % lower enrollment under the Gaussian measure. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-

similarity observations are also retained by setting $\widetilde{\log Q_p^k} = 0$ when $Q_p^k = 0$ and including $D_p^k \times \text{Post}_t$ separately. The coefficient on this indicator is not reported. Gaussian similarity is strictly positive in the exposed-market sample, so the corresponding indicator is absorbed by the fixed effects. All regressions absorb program, Broad-field $\times$ year, and pre-treatment region $\times$ year fixed effects. Standard errors clustered by pre-treatment market are in parentheses. Event studies reject the absence of differential pre-treatment trends, so estimates are interpreted as descriptive rather than causal.

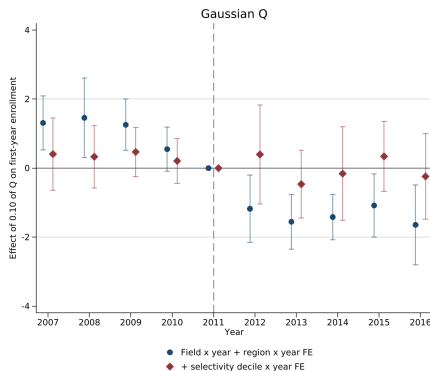
*** $p < 0,01$.

Within a market, entrants sit in a given PSU range, so Q_p^k is largely a function of the incumbent's own selectivity S_p : it falls monotonically from the lowest to the highest S_p decile (Triangular 0.35 to 0.04; Gaussian 0.68 to 0.17). S_p and S_p^2 explain 25 % (Triangular) and 47 % (Gaussian) of its variance, and 59 % and 77 % with market fixed effects.

Fixed effects	Levels		Logs		Event study: pretrend p	
	Triangular	Gaussian	Triangular	Gaussian	Triangular	Gaussian
Field $\times$ year + region $\times$ year	-2.381*** (0.491)	-2.241*** (0.487)	-0.067*** (0.020)	-0.142*** (0.035)	0.021	0.008
+ selectivity decile $\times$ year	-0.318 (0.446)	-0.309 (0.450)	-0.007 (0.017)	-0.048* (0.028)	0.785	0.631
+ $S_p, S_p^2 \times$ year	-0.384 (0.480)	0.039 (0.501)	-0.017 (0.016)	-0.075** (0.029)	–	–
Market $\times$ year	-2.791*** (0.564)	-2.570*** (0.570)	-0.069*** (0.019)	-0.149*** (0.036)	0.019	0.005
+ selectivity decile $\times$ year	-0.449 (0.480)	-0.341 (0.542)	-0.012 (0.017)	-0.055* (0.031)	0.756	0.236

Notes: Broad-field markets with entrant presence ($M_m > 0$); 591 programs, 36 markets. All regressions include program fixed effects. The first row reproduces the [main similarity table](#). Selectivity deciles are deciles of S_p , the incumbent's 2009–2011 mean PSU score, across programs. Levels: effect of a 0.10 increase in Q ; logs: $\log \widehat{Q} \times Post$ with $\mathbf{1}\{Q > 0\} \times Post$. Event-study columns report the joint 2007–2010 pretrend p -value of the levels event study. SE clustered by pre-treatment market. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

Conditional similarity: event studies with selectivity trends ← Table



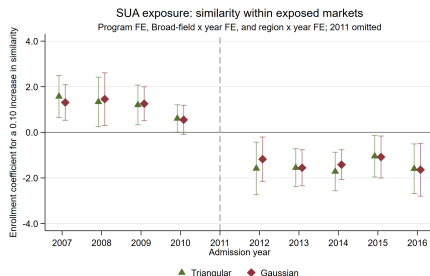
Notes: Levels event studies of first-year enrollment on $10 \times Q_p^k \times \mathbf{1}\{t = s\}$, 2011 omitted, with program, Broad-field $\times$ year and region $\times$ year fixed effects (blue), and additionally selectivity-decile $\times$ year fixed effects (red). 95 % confidence intervals; SE clustered by pre-treatment market. Under the base specification the coefficient declines steadily from 2007, so the post-2012 gap continues a pre-existing trend. Once selectivity trends are absorbed the path is flat before and after 2012 (pretrend $p = 0.785$ and 0.631).

Conditional similarity: heterogeneity by incumbent selectivity

	PSU < 550		PSU 550–649		PSU $\geq$ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
<i>Conditional similarity Q</i>						
Coefficient	-1,95*	-3,26	-2,20***	-2,17***	-5,70	-3,48
	(0,93)	(1,96)	(0,48)	(0,37)	(4,24)	(3,15)
F-statistic	4.40	2.77	21.27	34.14	1.80	1.22
Observations	836		2,932		1,038	
Programs	104		339		128	
Clusters	15		28		16	

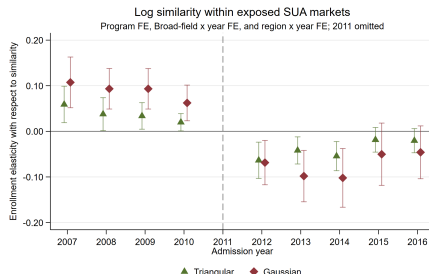
Notes: Standard errors clustered by pre-treatment market in parentheses. All regressions include program and market $\times$ year FE and are restricted to exposed markets. Coefficients correspond to a 0.10 increase in conditional similarity. The same split for SUA exposure is in the [main text](#). * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Levels



Exposed markets: economically meaningful zero similarity retained.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0209, Gaussian = 0.0078.

Logs

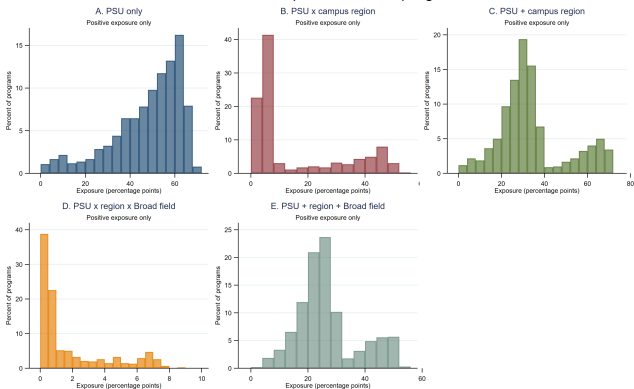


Exposed markets: Q = 0 retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0348, Gaussian = 0.0001.

Notes: Coefficients are relative to 2011. All models absorb program, Broad-field $\times$ year, and pre-treatment region $\times$ year fixed effects; standard errors are clustered by pre-treatment market. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-similarity observations are

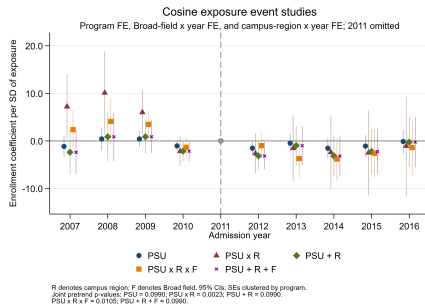
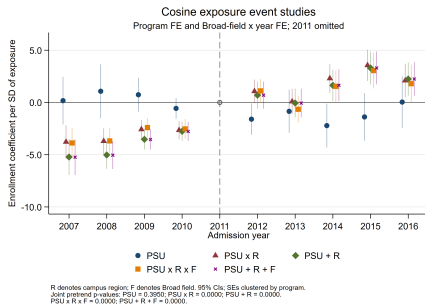
also retained by setting $\log Q_p^k = 0$ when $Q_p^k = 0$ and including $\mathbf{1}\{Q_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately in each event year; these indicator coefficients are not plotted. Observations with zero first-year enrollment are excluded from the log specification. Joint pretrend p-values are 0.0209 and 0.0078 for the triangular and Gaussian measures in levels, and 0.0348 and 0.0001 in logs. All four tests reject the absence of differential pre-treatment trends at the 5% level; estimates are therefore interpreted as descriptive associations rather than causal effects.

Cosine exposure distributions One observation per incumbent program



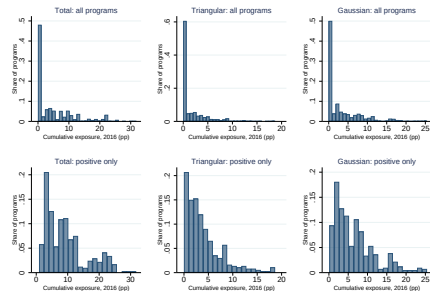
Positive exposure only. Exposure measured in percentage points; each panel uses its own x-axis scale.

- Programs with zero exposure are excluded, allowing the shape of the positive part of each distribution to be seen more clearly.
- The multiplicative $\text{PSU} \times \text{campus region} \times \text{Broad field}$ measure remains concentrated at relatively low exposure values even after conditioning on positive exposure.



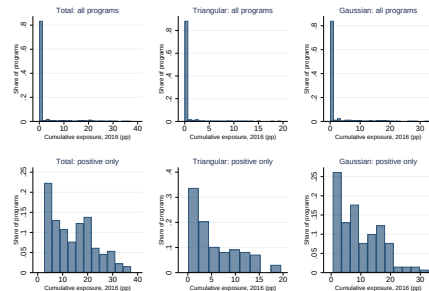
- Without campus-region $\times$ year FE, PSU only does not reject joint pretrends ($p = 0,395$), but all four measures that incorporate campus region do ($p < 0,001$).
- Adding campus-region $\times$ year FE improves the diagnostic for PSU and the additive measures (all $p = 0,099$), but the multiplicative measures still reject: PSU $\times$ region ($p = 0,0023$) and PSU $\times$ region $\times$ field ($p = 0,0105$).
- Post-2012 estimates become mostly negative and imprecise once campus-region $\times$ year FE are included. The min-10 sample is not shown and delivers virtually the same conclusions.
- Overall, the cosine results are best interpreted as descriptive rather than causal.

ISCED-97



Incumbent programs, cumulative exposure in 2016; trimmed at p99. ISCED-97-field markets.

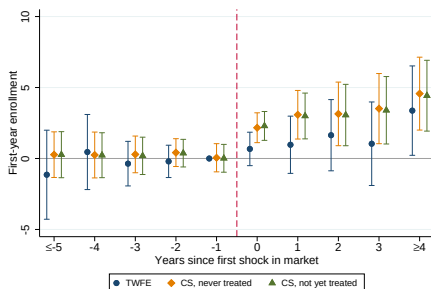
Generic



Incumbent programs, cumulative exposure in 2016; trimmed at p99. Generic-field markets.

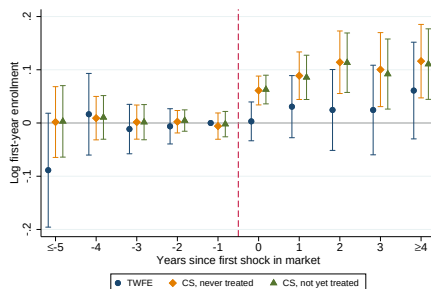
Narrower field definitions generate more markets and more zero-exposure programs: 83 % under the Generic definition.

First-year enrollment



Pretrend p-values: TWFE 0.348; CS never-treated 0.080; CS not-yet-treated 0.081.
TWFE: program, field-year, region-year FE, own shocks; $t = -1$ omitted. CS: balanced panel.

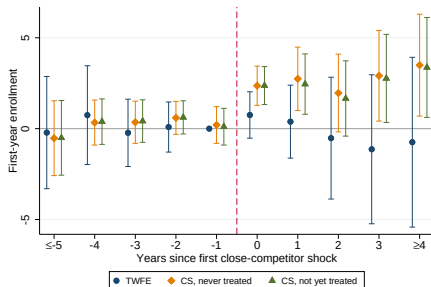
Log first-year enrollment



Pretrend p-values: TWFE 0.124; CS never-treated 0.011; CS not-yet-treated 0.012.
TWFE: program, field-year, region-year FE, own shocks; $t = -1$ omitted. CS: balanced panel.

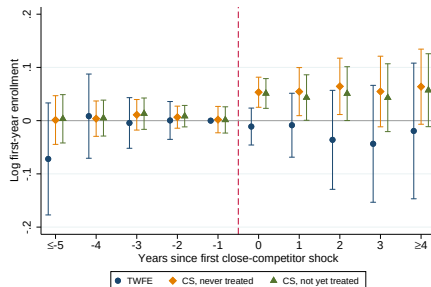
Notes: g_k is the first year with a shock at another program in k 's Broad-field $\times$ region market. TWFE absorbs program, field $\times$ year and region $\times$ year FE, controls for own shocks, omits $t = -1$ and clusters by market. Callaway & Sant'Anna (csdid) uses the balanced panel with never-treated controls (orange) or not-yet-treated plus never-treated controls (green). 95 % confidence intervals.

First-year enrollment



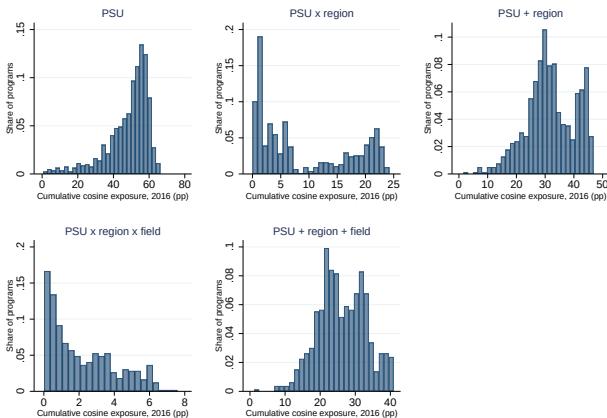
Pretrend p-values: TWFE 0.539; CS never-treated 0.000; CS not-yet-treated 0.000.
TWFE: program, field-year, region-year FE, own shocks; t = -1 omitted. CS: balanced panel.

Log first-year enrollment



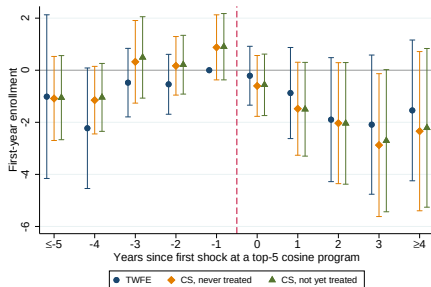
Pretrend p-values: TWFE 0.240; CS never-treated 0.000; CS not-yet-treated 0.000.
TWFE: program, field-year, region-year FE, own shocks; t = -1 omitted. CS: balanced panel.

Notes: g_k is the first year with a shock at a program k 's Broad-field market whose entrant PSU is within 50 points of k 's (the support of the triangular kernel). Estimators as in the previous slide.



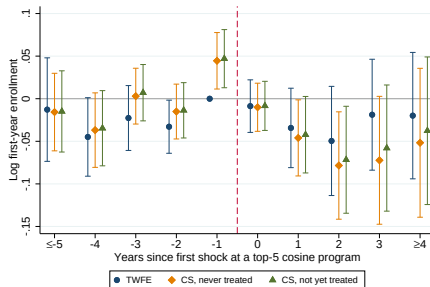
Incumbent programs, full sample. Programs with positive exposure only.

First-year enrollment



Pretrend p-values: TWFE 0.293; CS never-treated 0.021; CS not-yet-treated 0.021.
TWFE: program, field-year, region-year FE, own shocks; t = -1 omitted. CS: balanced panel.

Log first-year enrollment



Pretrend p-values: TWFE 0.150; CS never-treated 0.068; CS not-yet-treated 0.069.
TWFE: program, field-year, region-year FE, own shocks; t = -1 omitted. CS: balanced panel.

Notes: g_k is the first year with a shock at one of the five programs with the highest PSU-bin cosine similarity to k , anywhere in the country. TWFE clusters by program. Estimators as in the market-shock slide.